

Computational Fluid Dynamics Analysis of Heat Sink With Phase Change Material

A project report submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology
in
Mechanical Engineering Discipline

by
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Abstract

This project, "Computational Fluid Dynamics Analysis of Heat Sink With Phase Change Materials," offers a thorough investigation into optimizing the thermal performance of heat sinks used for electronic cooling tasks. The main emphasis is on using Phase Change Materials (PCMs) to improve heat dissipation efficiency under different thermal loads with extended fin configurations and PCM compositions.

The study simulates and examines the behavior of heat sinks with different heat input parameters and extended fins with PCM compositions using ANSYS's Computational Fluid Dynamics (CFD) capabilities. By effectively controlling heat inputs through the PCMs' melting and solidification processes, the goal is to find the best design that optimizes thermal performance.

The study investigates the effects of different heat inputs and boundary conditions on the efficiency of the heat sink in addition to thermal performance studies. In order to comprehend the temperature distribution and phase change dynamics within the heat sink, the study evaluates heat sink models and compares their outcomes. The results are presented using a variety of graphs, including outlet temperature vs. time, liquid percent vs. time, PCM temperature vs. time, and temperature and liquid fraction contours.

Certificate

It is certified that the work contained in the project report titled, “**Computational Fluid Dynamics of Heat Sink With Phase Change Material**”, submitted by **Nishant Kumar (Roll No: 21BME029)** and **RajKumar Singh (Roll No: 21BME040)** has been carried out under my supervision and guidance.

To the best of my knowledge, the matter embodied in the report has not been submitted elsewhere to any other university/institute for the award of any other degree.

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We would also like to thank the Mechanical department of IIITDMJ for providing necessary resources and facilities, which were crucial for the progress of this project.

My deepest gratitude goes to my family and friends for their continuous encouragement and support. Their patience and understanding have been a source of strength throughout this journey.

Finally, we extend our appreciation to all those who, directly or indirectly, supported and motivated me to bring this project to fruition. This work is a testament to the collective effort and dedication of everyone involved.

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INDEX

SR.NO	TITLE	PG NO.
1.	CHAPTER 1 1.1 INTRODUCTION 1.2 PROBLEM STATEMENT	5
2.	CHAPTER 2 2.1 LITERATURE REVIEW 2.2 OBJECTIVE 2.3 PROJECT PLAN	6
3.	CHAPTER 3 3.1 COMPUTATIONAL FLUID DYNAMICS SETUP 3.1.1 CFD MODEL 3.1.2 GRID GENERATION 3.1.3 CFD ANALYSIS 3.1.4 SYSTEM PARAMETERS 3.1.5 MODEL PARAMETERS 3.1.6 BOUNDARY CONDITIONS	11
4.	CHAPTER 4 4.1 RESULT AND DISCUSSIONS 4.1.2 PIN FIN CONFIGURATION AT 25W 4.1.2 PIN FIN CONFIGURATION AT 30W 4.1.3 PIN FIN CONFIGURATION AT 35W 4.2 COMPARISON BETWEEN THE PIN FIN CONFIGURATION AT DIFFERENT POWER LEVEL OF 25W, 30W AND 35W	18
5.	CFD MODEL VALIDATIONS	38
6.	CONCLUSIONS	41
7.	REFERENCES	42

CHAPTER 1

1.1 Introduction

The rising need for high-performance electronic gadgets in recent years has resulted in compact systems producing more heat and having higher power densities. Effective heat control is now essential to guaranteeing these equipment's dependability and durability. With their ability to efficiently disperse heat from electronic components, heat sinks are one of the most promising solutions in thermal management. In terms of improving heat dissipation capabilities, heat sinks—which incorporate several thermal management techniques—have demonstrated considerable promise among the numerous types of heat sinks.

In order to assess a heat sink's thermal performance under varied circumstances, this research employs ANSYS software to perform a Computational Fluid Dynamics (CFD) analysis. The main goals are to examine the heat sink's efficiency under various conditions and the contribution of Phase Change Materials (PCMs) to improving its thermal performance. The latent heat storage capacity of PCMs is well known, and it can be used to stabilize the temperature of electronic components by absorbing and releasing heat during phase transitions.

The study includes a thorough analysis of the PCMs' melting and solidification processes inside the heat sink. Contour plots produced by ANSYS simulations are used to visualize temperature distribution and phase change processes, offering important information about the heat sink's thermal behavior under various operating conditions.

1.2 Problem Statement

Higher power densities brought about by the development of electronic devices result in a considerable amount of heat output, which needs to be efficiently controlled to avoid overheating and maintain device dependability. Despite their relative effectiveness, traditional heat sinks are becoming less and less capable of managing the thermal demands of contemporary high-performance electronics. Creating a heat sink design that can effectively dissipate heat and sustain ideal operating temperatures in a range of thermal situations is the difficult part.

In order to overcome this difficulty, this study examines the functionality of a heat sink that incorporates Phase Change Materials (PCMs) for improved thermal control. The main challenge is figuring out how different heat inputs with extended fins impact the heat sink's efficiency and comprehending how PCMs' melting and solidification processes contribute to temperature stabilization. With the use of ANSYS software and thorough CFD research, this study seeks to determine the best setups and circumstances for optimizing heat dispersion and enhancing the heat sink's overall thermal performance.

Chapter 2

2.1 Literature Review

[1] In this paper Ali et al. (2018), including researchers Hafiz Muhammad Ali, Muhammad Junaid Ashraf, Ambra Giovannelli, and colleagues, experimentally analyzed triangular, rectangular, and circular pin-fin heat sinks integrated with six PCMs (paraffin wax, RT-54, RT-44, RT-35HC, SP-31, n-eicosane) for electronics cooling. Testing under 5–8 W power inputs, triangular pin-fins demonstrated superior thermal performance due to lower cross-sectional area and higher fin density, extending operational time. PCMs with high latent heat (e.g., RT-35HC, SP-31) optimized energy storage, while Stefan number analysis guided PCM selection. The study, conducted at the University of Engineering and Technology, Taxila, with international collaborators, highlights geometric and material synergies for efficient passive thermal management in compact electronics.

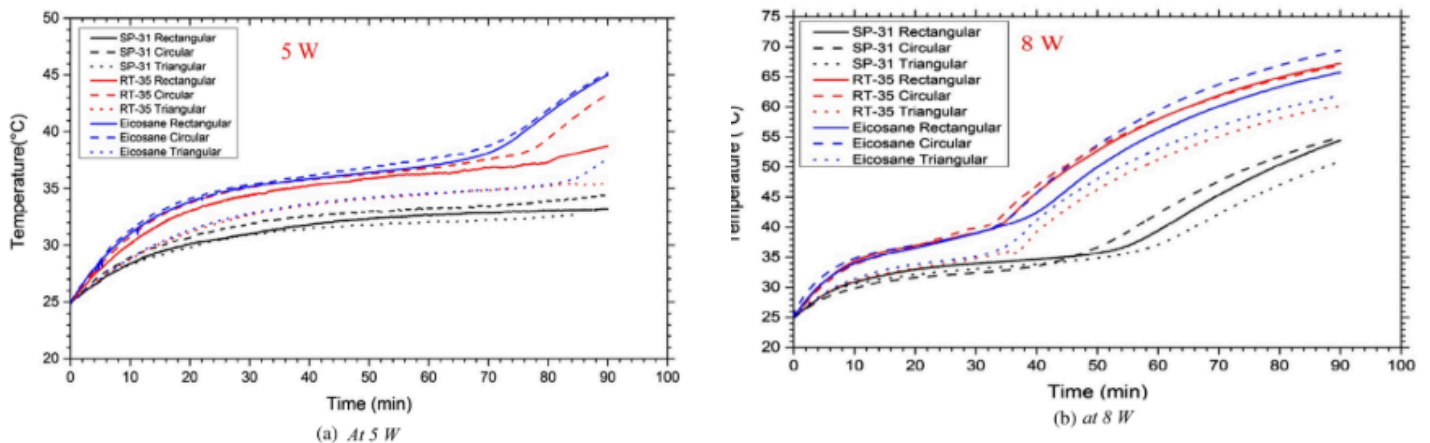
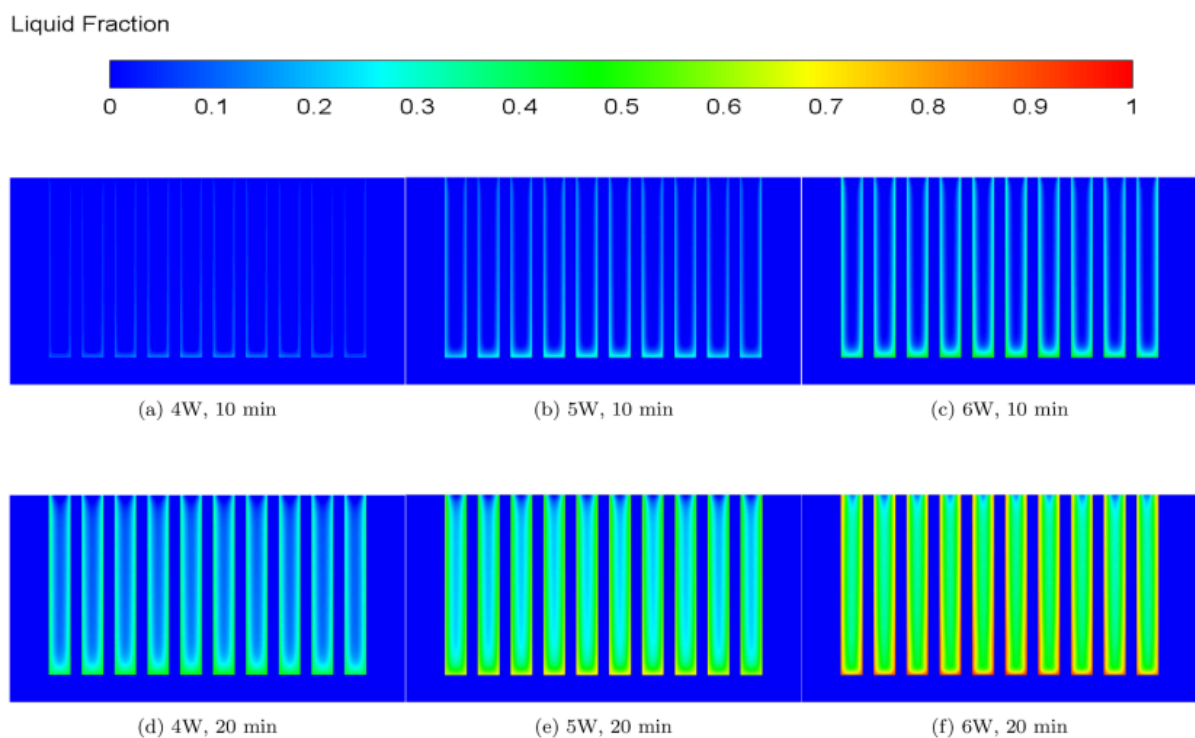


Fig. Comparison between rectangular, round and triangular pin-fins for high melting PCMs.

[2] In their paper, V. Shatikian, G. Ziskind, and R. Letan studied how a phase-change material (PCM) melts inside a heat sink with a heated base and vertical fins. They used simulations to analyze the process, considering factors like heat conduction in the fins, heat transfer and fluid flow in the melted PCM, volume changes during melting, and heat loss to the surroundings. Their research explored how different designs and conditions affect heat storage in the PCM. The findings highlight the importance of fins in spreading heat efficiently. This method can help improve the design of PCM-based cooling systems.

[3] In their paper, Adeel Arshad et al. investigate phase change material (PCM)-based finned heat sinks for passive cooling of electronics. Using n-eicosane PCM and aluminum fins (2 mm/3 mm thick, 9% volume fraction) in a 72×25 mm heat sink, transient ANSYS Fluent simulations analyze thermal performance under 4–6 W heating. The enthalpy-porosity method models PCM melting, with natural convection at the top and adiabatic sidewalls. Results show 3 mm fins maintain lower base temperatures despite longer melting times. Higher power accelerates PCM melting. Dimensionless correlations link liquid fraction and Nusselt number to Fourier, Stefan, and Rayleigh numbers. The study, led by Arshad, Jabbar, Sardari, Bashir, Faraji, and Yan, highlights PCM-fin synergies for electronics thermal management.



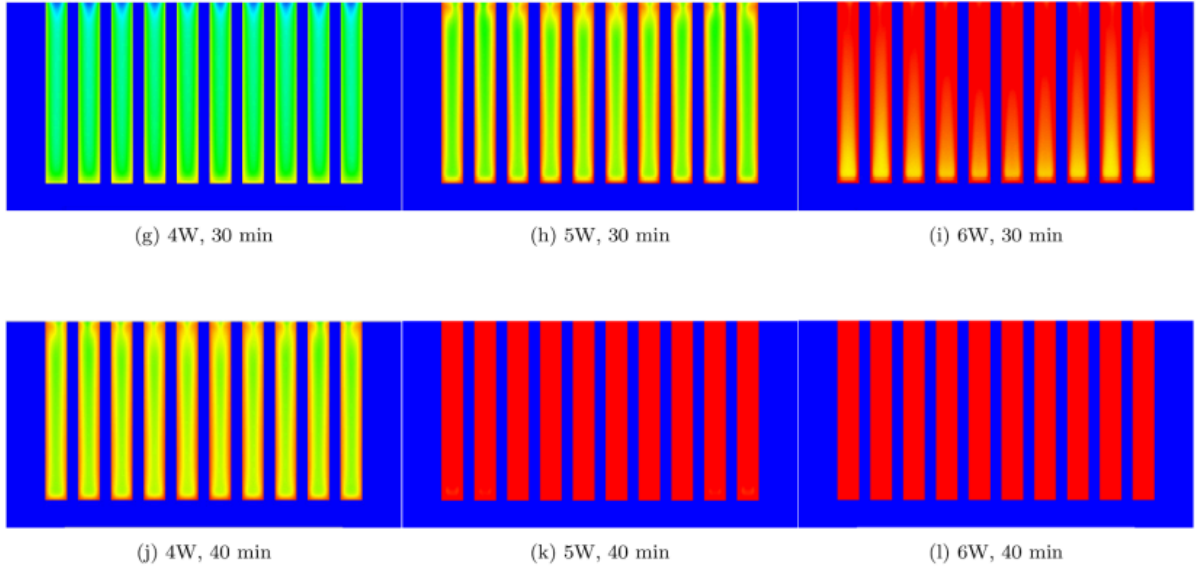
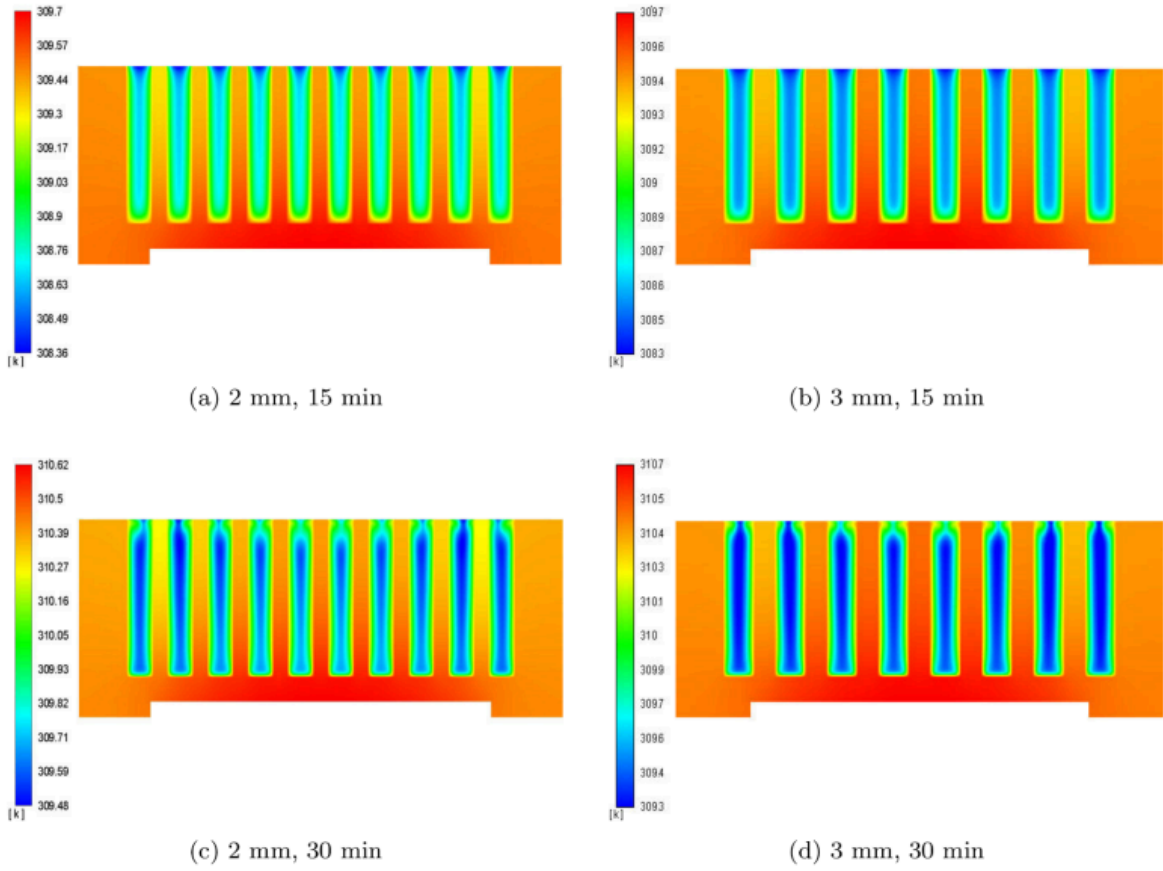


Fig. Variation of liquid fraction of 2 mm fin thickness at various time periods and input heat fluxes



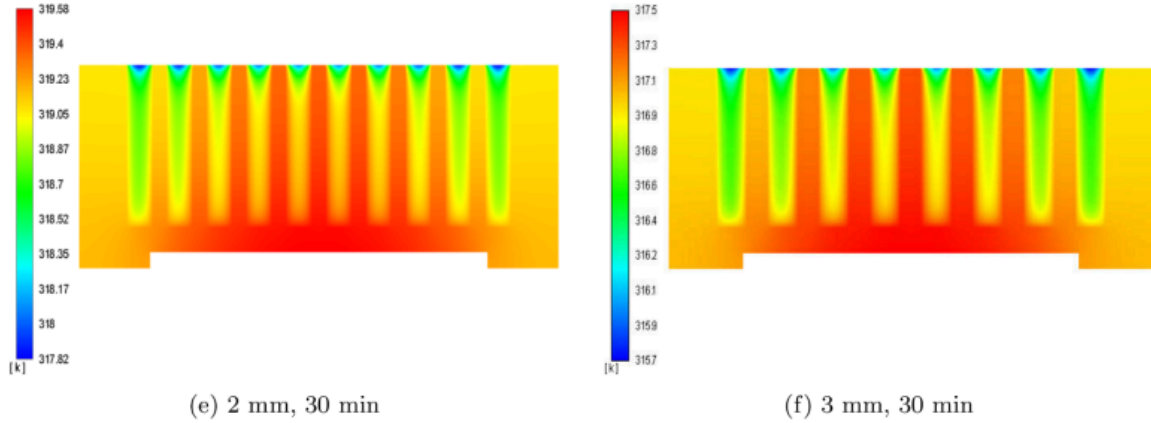


Fig. Comparison of isotherms contours at various time period of 2 mm and 3 mm fin thickness cases at power level of 5 W

[4] In this paper, X. Duan and G.F. Naterer studied how phase change materials (PCMs) can be used to manage heat in electric vehicle battery modules. They tested whether PCMs work well for this purpose by analyzing melting and solidification processes and sharing new experimental data.

To simulate a battery cell, they used a heater and tested two PCM cooling setups, A PCM container around the heater and a PCM jacket wrapped around the heater. Both designs successfully kept the heater's temperature within a safe range. The researchers also looked at how different heating speeds and outside temperatures affect the cooling performance. Their findings show that PCMs can effectively control battery temperatures in electric vehicles.

2.2 Objective

The main goal of this project is to thoroughly analyze a heat sink using CFD (Computational Fluid Dynamics) simulations to study how well it manages heat under different operating conditions. The key aims are:

(1) Design and Model Development: To design a heat sink with extended fins which uses Phase Change Materials (PCMs) and build a detailed CFD simulation model in ANSYS.

(2) Thermal Performance Analysis: To analyze the thermal performance of the heat sink under different heat inputs and assess its ability to dissipate heat effectively.

- (3) PCM Behavior Evaluation:** To observe the melting and solidification processes of the PCMs within the heat sink and their impact on the thermal management capabilities.
- (4) Contour Plot Visualization:** To Create and analyze liquid fraction and temperature maps (contour plots) to visualize heat spread and PCM melting/solidifying patterns in the heat sink.
- (5) Performance Optimization:** To Find the best heat sink setup and operating conditions to boost cooling performance, with real-world tips for high-power electronics.
- (6) Comparison and Validation:** To test the advanced heat sink against standard designs and check if simulation results match real-world tests, confirming the findings are accurate.

2.3 Methodology

2.3.1 Literature Review

The project started by examining current knowledge about heat sinks and cooling technologies, with emphasis on advanced systems incorporating PCMs. This review deepened our understanding of thermal processes and how PCMs boost heat dissipation in electronics.

2.3.2 Model Designing

Using SolidWorks 2020, we developed several heat sink designs featuring PCM compartments. We will vary the heat inputs and PCM configuration to enhance heat dissipation, always balancing thermal performance with structural durability and production feasibility.

2.3.3 Simulations

We will analyze the simulation results in ANSYS CFD Post, tracking critical parameters like

PCM temperature, air outlet temperature, and PCM melt percentage. This data will help us to evaluate the heat sink's cooling performance across different operating scenarios and validate the design effectiveness.

2.3.4 CFD Post Data Collection

We will analyze the simulation results in ANSYS CFD Post, tracking critical parameters like PCM temperature, air outlet temperature, PCM melt percentage, temperature contours and liquid fraction contours. This data will help us to evaluate the heat sink's cooling performance across different operating scenarios and validate the design effectiveness.

Chapter 3

3.1 Computational Fluid Dynamics Setup

3.1.1 CFD Model

The geometric and operational parameters for this study were determined based on prior experimental research. A three-dimensional CFD model of the hybrid heat sink was developed in SolidWorks with overall dimensions of 25 mm × 72 mm × 72 mm. The design features a square plate to enhance heat transfer. Cylindrical fins measuring 20 mm in length and 3 mm in diameter were incorporated, spaced 5.3 mm apart to ensure adequate thermal flow development before the air exits the test section. All walls, except the absorber plate (insulated), were modeled as smooth and conductive for smooth cooling of the finned structure and PCM components.

Dimensions of Heat Sink are mentioned in given below table :

Components	(mm)
Heat Sink	75 x 75 x 25
Fin's Diameter	3
Fin's Height	20
Fin's Gap	5.3

Heat Sink Geometry

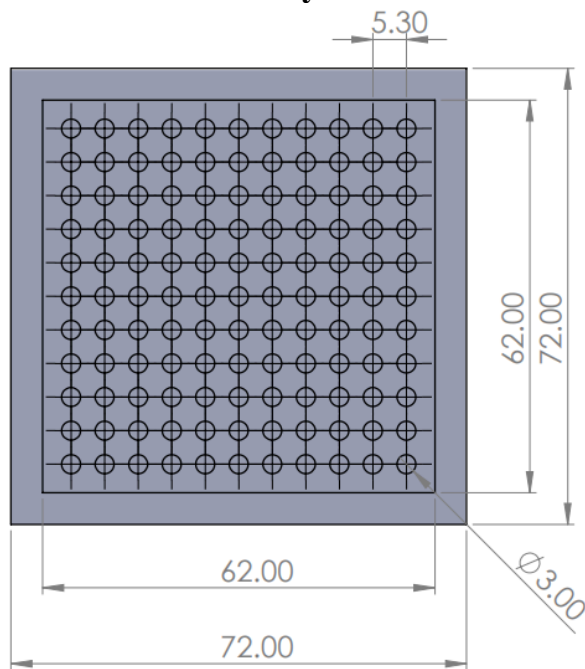


Fig: Top view of heat sink

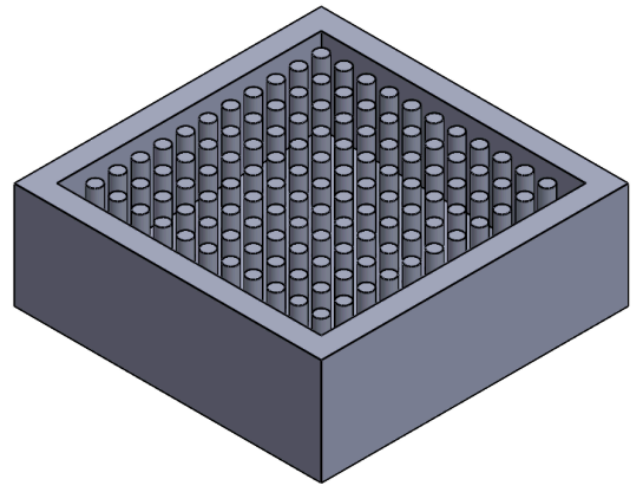


Fig: Isometric view of heat sink

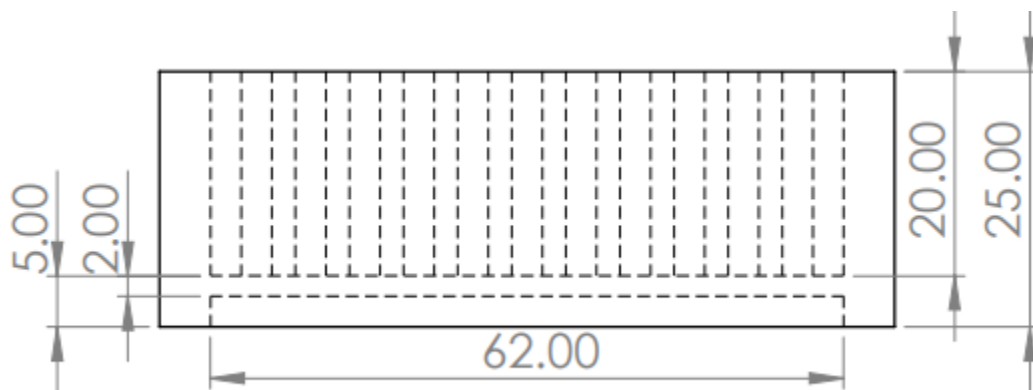


Fig: Front view of heat sink

All dimensions are in mm

3.1.2 GRID GENERATION

The mesh generation process was carried out using the ANSYS Meshing module. To properly capture the flow behavior, the three-dimensional computational domain was

segmented into multiple distinct components. A structured 3D mesh was then generated by applying standard sizing controls along with adaptive meshing techniques to optimize the grid resolution.

The various meshing pattern applied on the model are :

- 1) Sweep Method on Fins with 25 numbers of divisions.
- 2) Sweep Method on Side Walls with 25 divisions .
- 3) Sweep Method on Pcm with 23 divisions .
- 4) Edge sizing on various edges with different parameters such as 19,22 divisions .
- 5) Face Meshing on the base plate of the heat sink .
- 6) Multizone Meshing on the base aluminium body of the heat sink.

Meshing breaks down the heat sink geometry into small elements that computers can analyze. Using proper meshing methods is vital for getting accurate and reliable simulation results without unnecessary computational costs.

Number of Nodes	Number of Elements
120740	95175

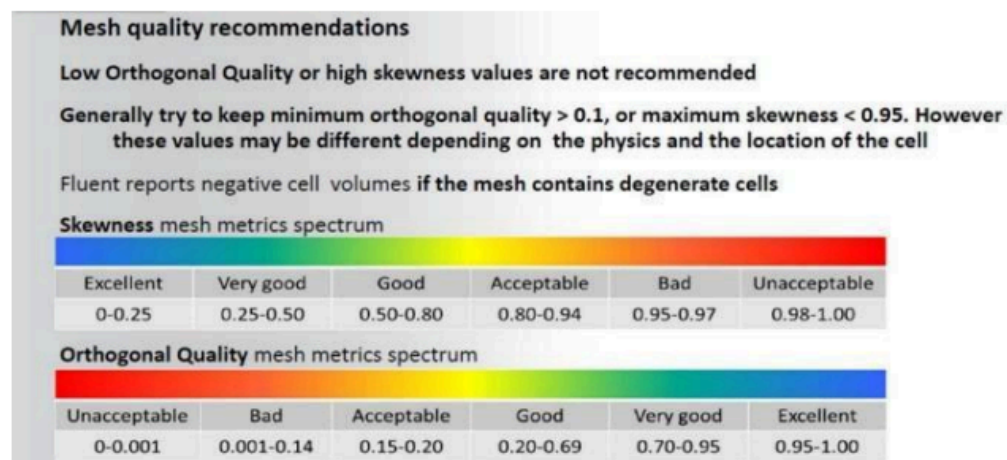


Fig: Mesh quality

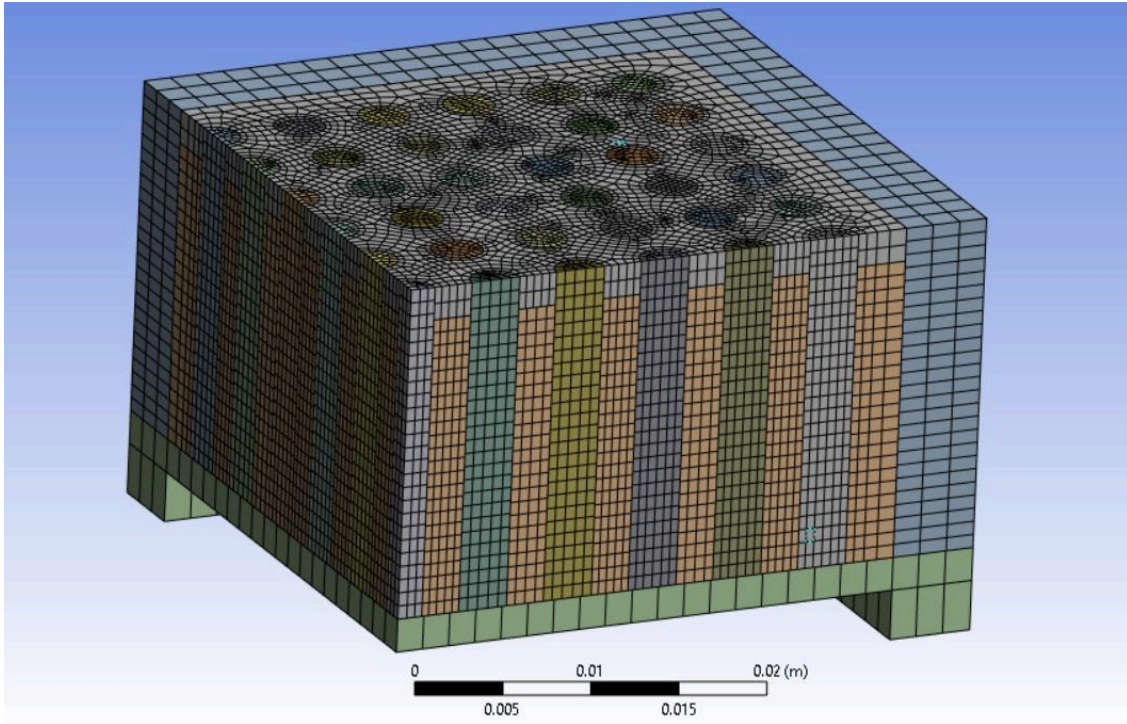


Fig: mesh

3.1.3 CFD ANALYSIS

Since the heat sink is designed to handle moderate temperature increases in electronics, the fluid density and thermal conductivity variations remain small enough to have negligible effects on the results. This approach is validated by existing literature, allowing air properties along the heat sink to be treated as constant at the bulk mean temperature.

For electronic devices with moderate heat generation, the heat sink's performance isn't significantly affected by changes in air density or thermal conductivity. Research supports treating these air properties as constant along the heat sink's length at average operating temperatures.

The analysis assumes incompressible airflow under time-varying conditions, with minimal changes in air properties. Air enters the system at ambient temperature and flows smoothly across surfaces (no-slip condition). Incorporating a phase-change material (PCM) with a 352-355K melting range significantly improves the heat sink's thermal management capabilities.

3.1.4 SYSTEM PARAMETERS

The geometric dimensions of heat sink with cylindrical extended fin's are mentioned in given below table:

System parameters	Range/base values (mm)
Fin's diameter	3
Base plate thickness	5
fins length	20
fins gap	5.3
Height of heat sink	25
Length / Width of heat sink	72

The n-eicosane phase change material is used as a testing material and properties of which are mentioned in given below table:

PCM properties	
Density(boussinesq)	790 kg/m ³
Viscosity	0.00355 kg/m-s
Solidus Temperature	308.5 K
Specific heat	2050 J/ kgK

Thermal conductivity	0.23 W/mK
Melting Heat	241000 J/kg
Liquidus Temperature	310.5 K

3.1.5 MODEL PARAMETERS

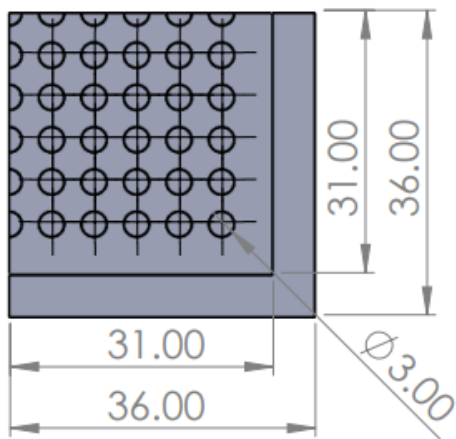


Fig: Top view of heatsink

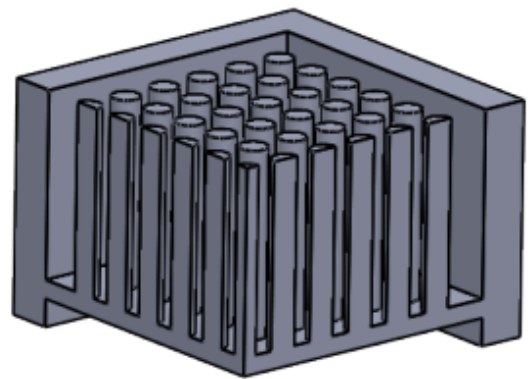


Fig: Isometric view of heatsink

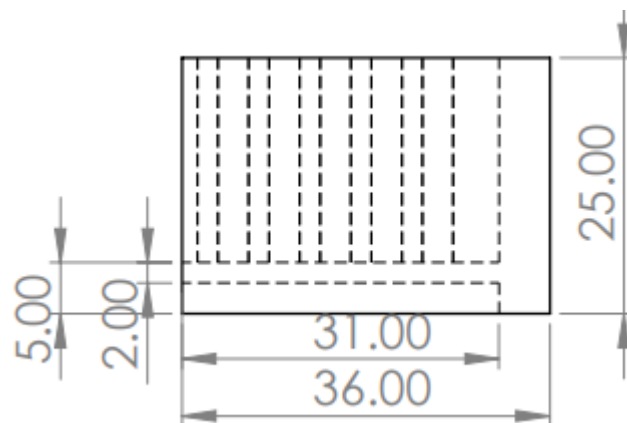


Fig: Front view of heatsink

3.1.6 BOUNDARY CONDITIONS

The study employs an aluminum heat sink featuring cylindrical fins, with phase change material (PCM) filling the inter-fin cavities. Thermal performance was evaluated under different power levels 25W, 30W and 35W applied to the base plate, examining forced convection scenarios. Given below table shows the material properties of the PCM and other components used in the simulations.

Material	Density(kg/m ³)	Specific Heat(J/kg K)	Thermal Conductivity (W/mK)
Aluminum	2719	871	202.4
Air	1.225	1006.43	0.0242
PCM	790	2050	0.23

Boundary	Conditions
Air-top Convection	20 W/m ² K
Fin-Top Convection	20 W/m ² K
Heat Input	25W, 30W and 35W
Side-wall convection	18 W/m ²
Wall-Top Convection	20 W/m ²

Chapter 4

4.1 RESULT AND DISCUSSIONS

4.1.1 THE PIN FIN CONFIGURATION AT 25 W

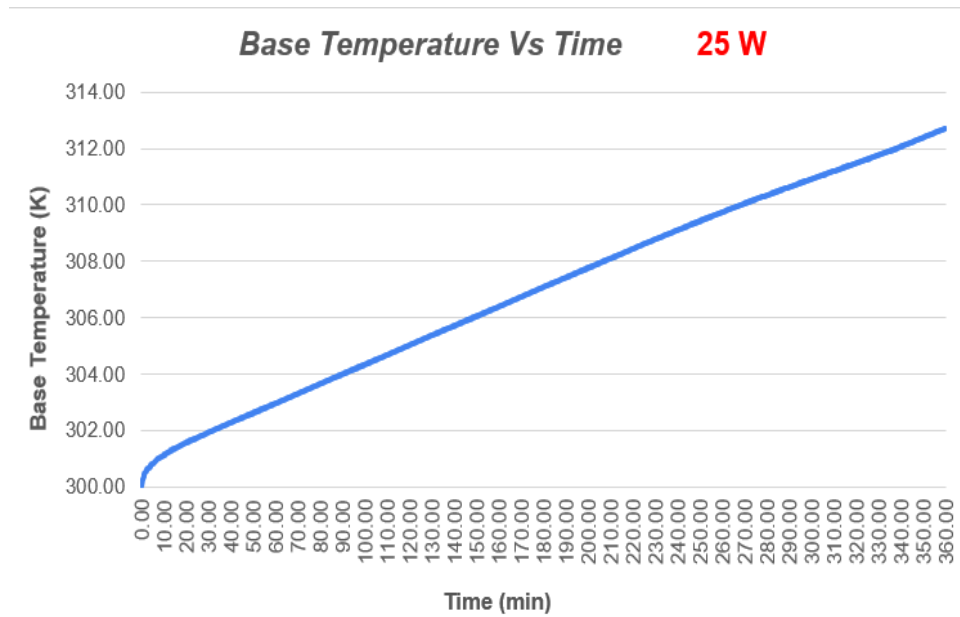


Fig: Base Temperature vs Time graph

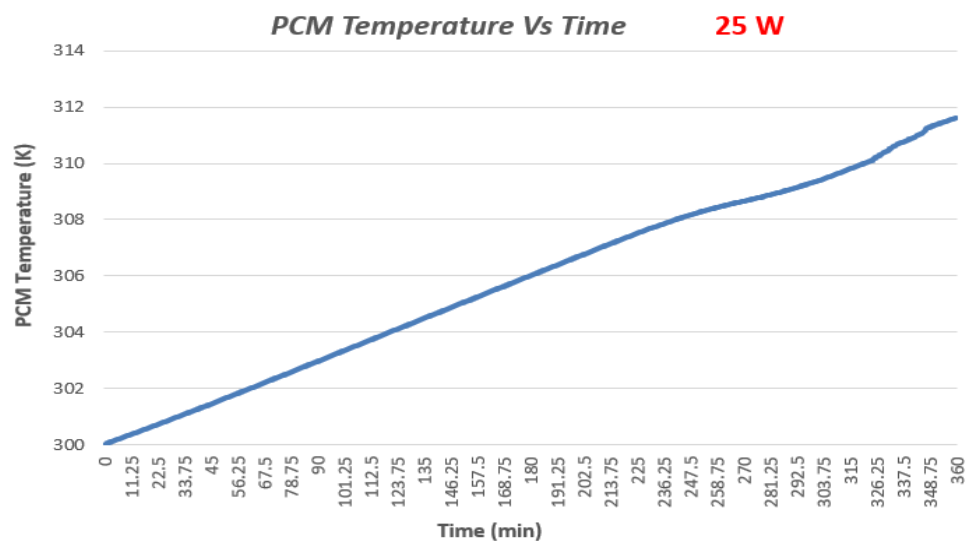


Fig: PCM Temperature vs Time graph

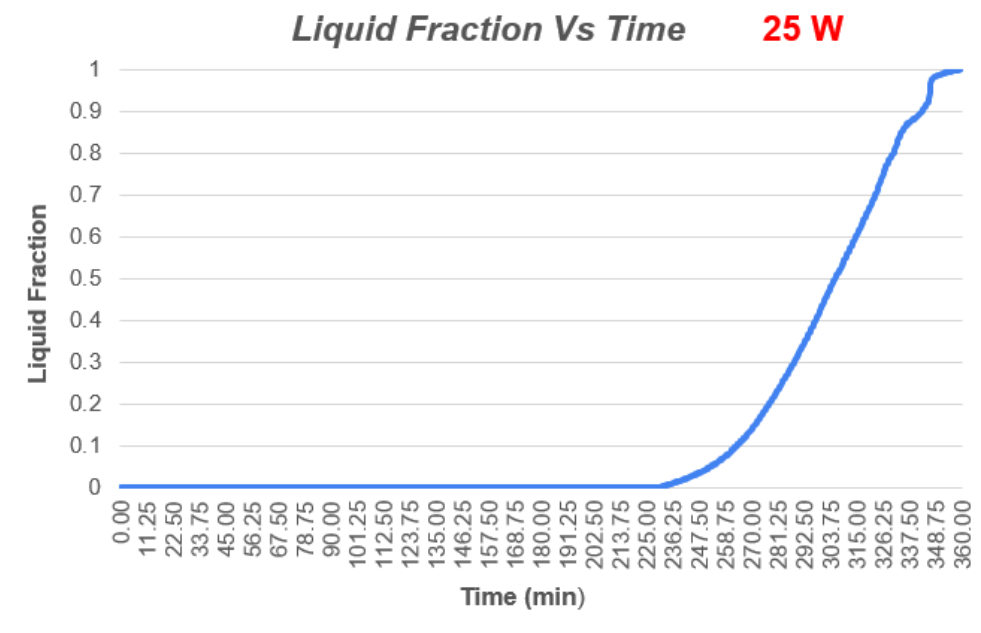
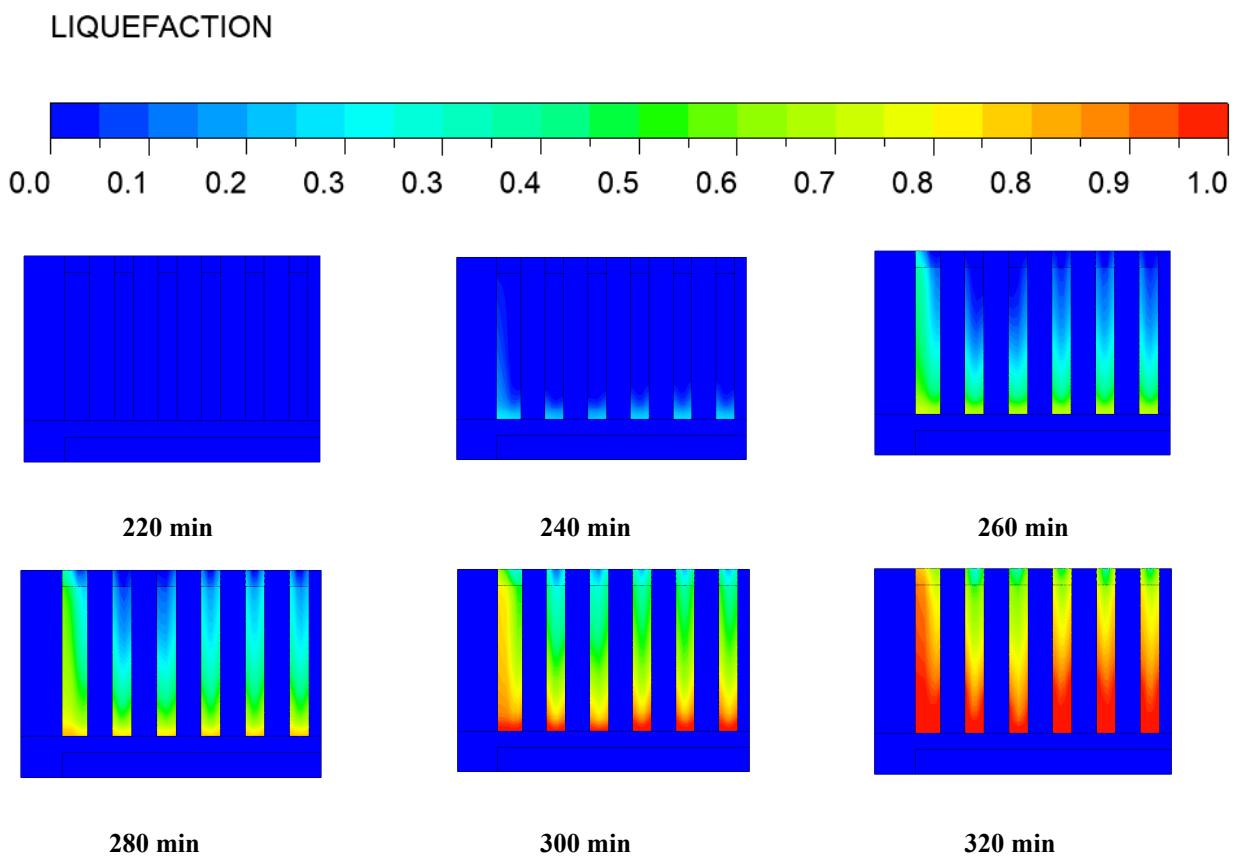
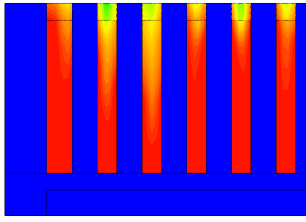


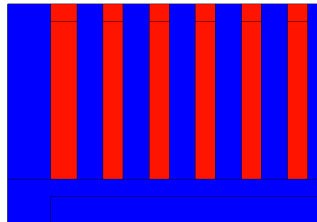
Fig: Liquid Fraction vs Time graph

4.1.1.1 Liquid Fraction Contours at power level of 25 W



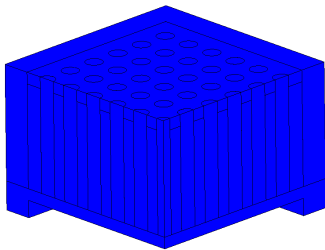


340 min

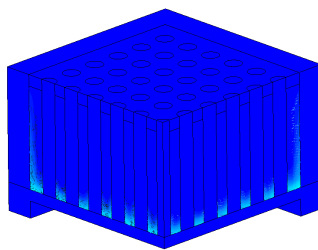


360 min

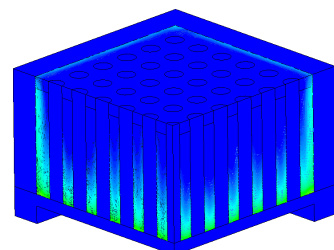
Fig. 2D view of liquid fraction contours at various time periods at power level of 25 W.



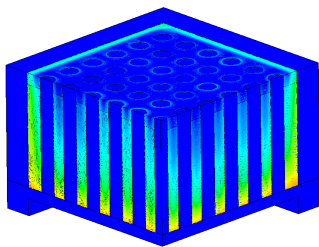
220 min



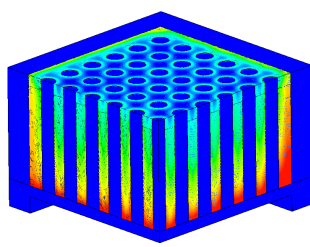
240 min



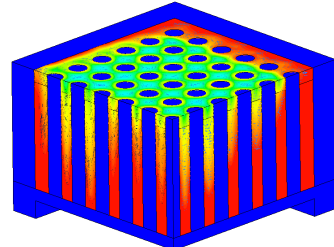
260 min



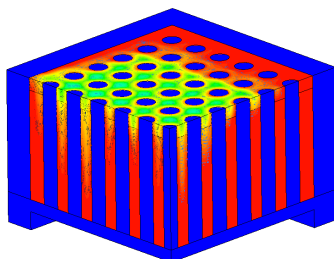
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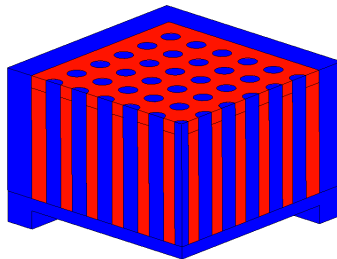
300 min



320 min



340 min



360 min

Fig. 3D view of liquid fraction contours at various time periods at power level of 25 W.

4.1.1.2 Temperature Contours at power level of 25 W

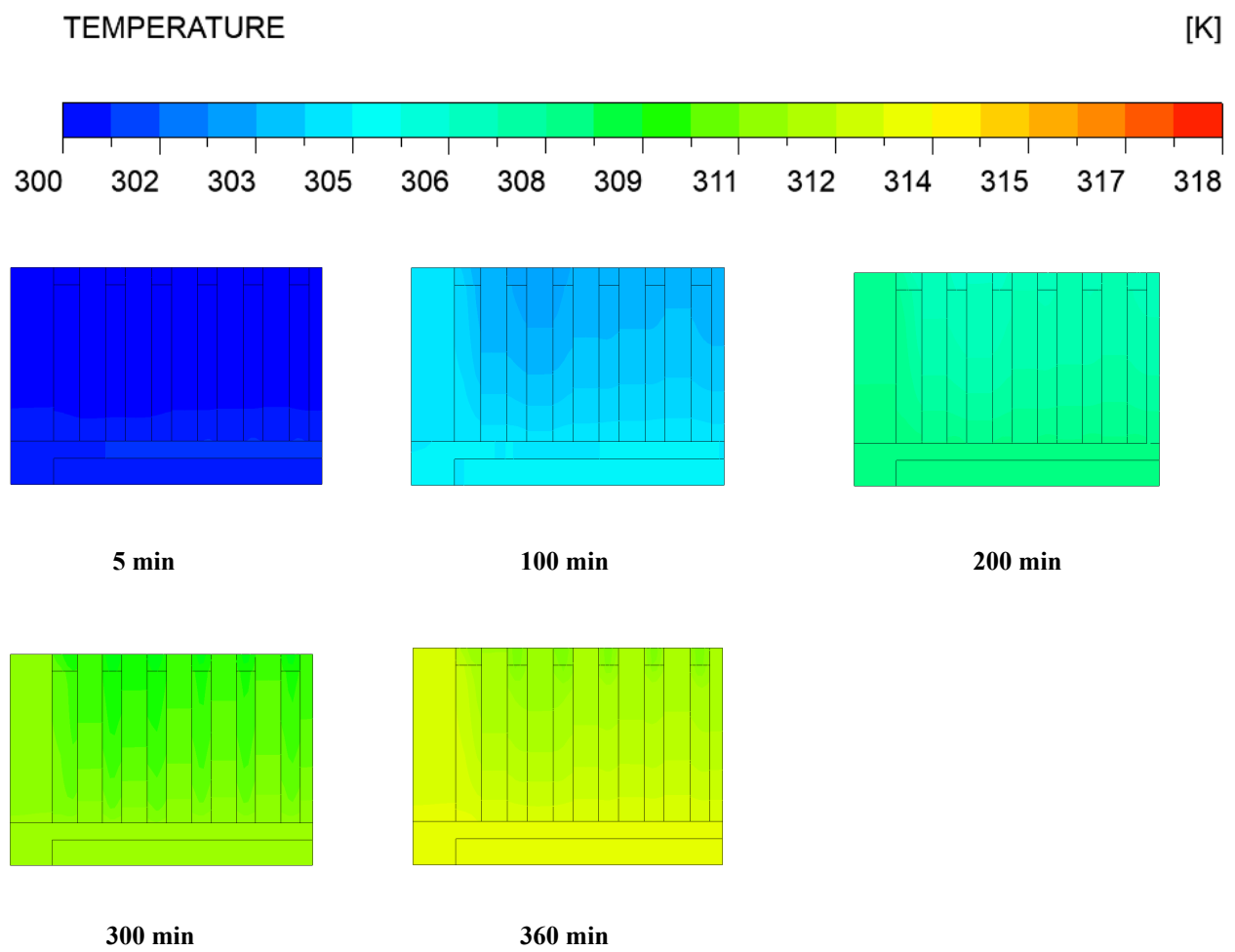
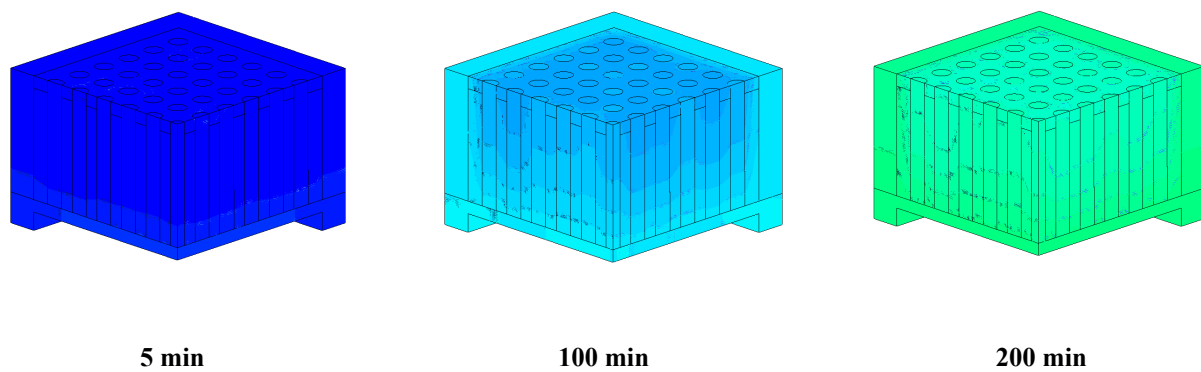
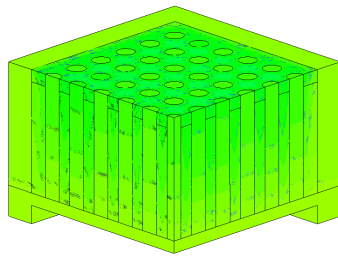
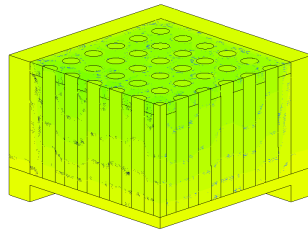


Fig. 2D view of temperature contours at various time periods at power level of 25 W.





300 min



360 min

Fig. 3D view of temperature contours at various time periods at power level of 25 W.

4.1.2 THE PIN FIN CONFIGURATION AT 30 W

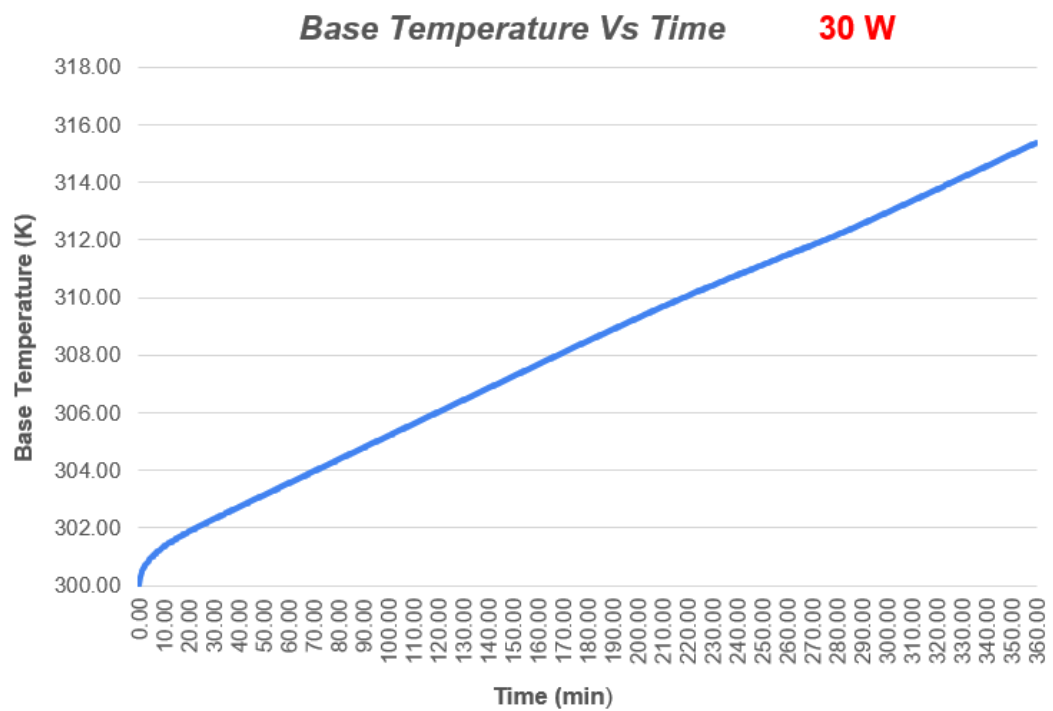


Fig: Base Temperature vs Time graph

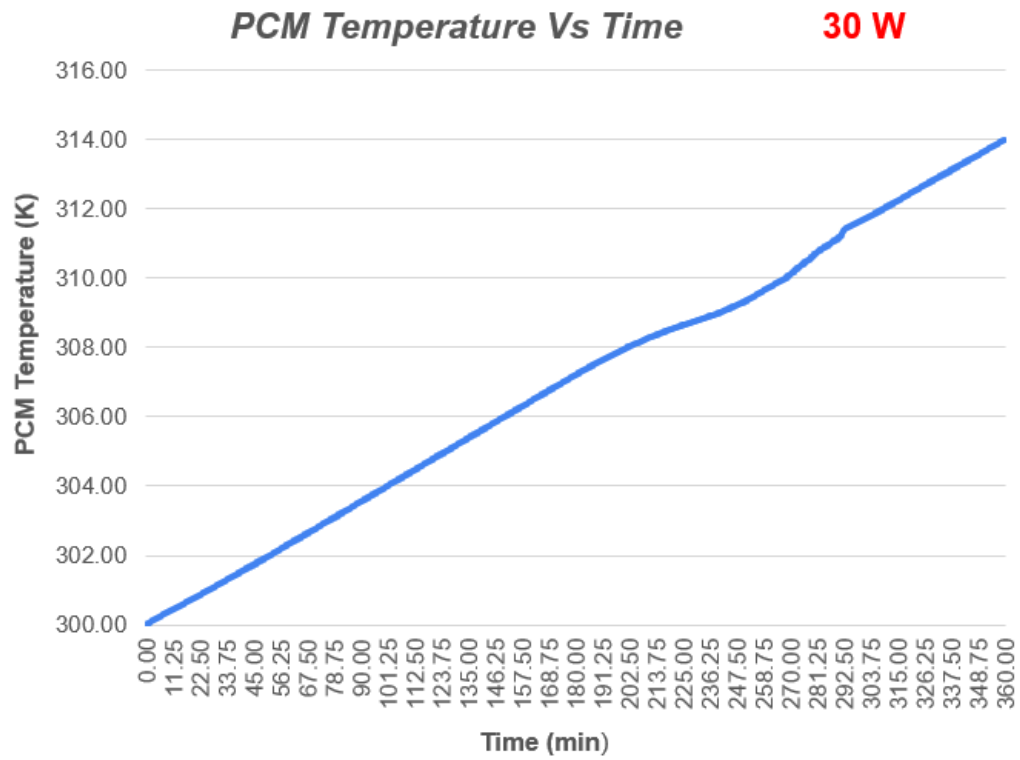


Fig: PCM Temperature vs Time graph

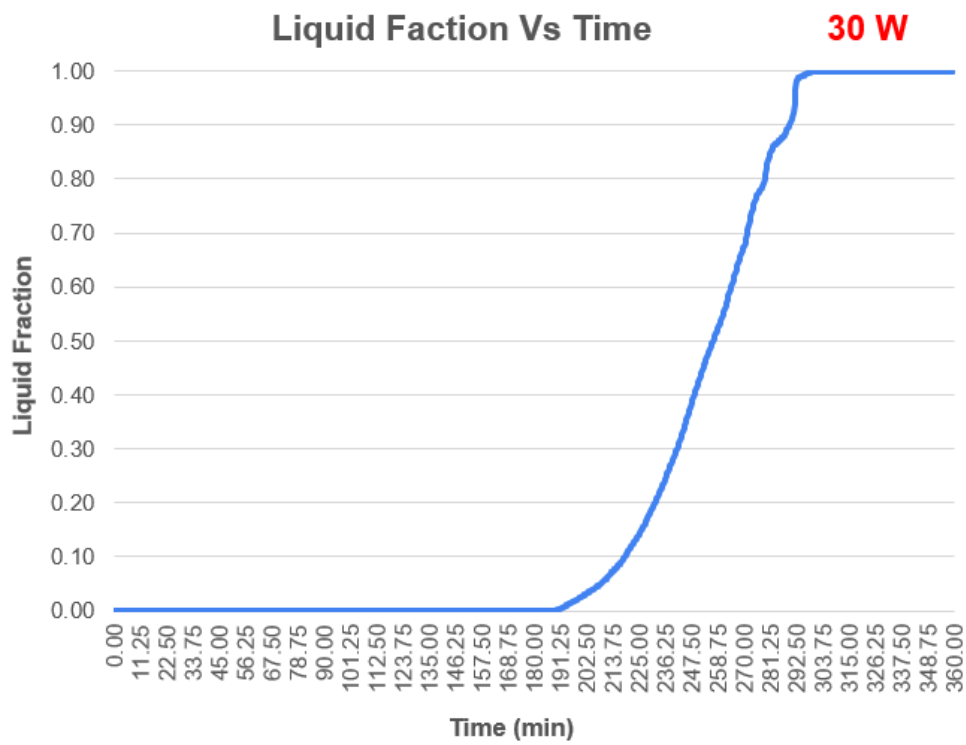


Fig: Liquid Fraction vs Time graph

4.1.2.1 Liquid Fraction Contours at power level of 30W

LIQUEFACTION

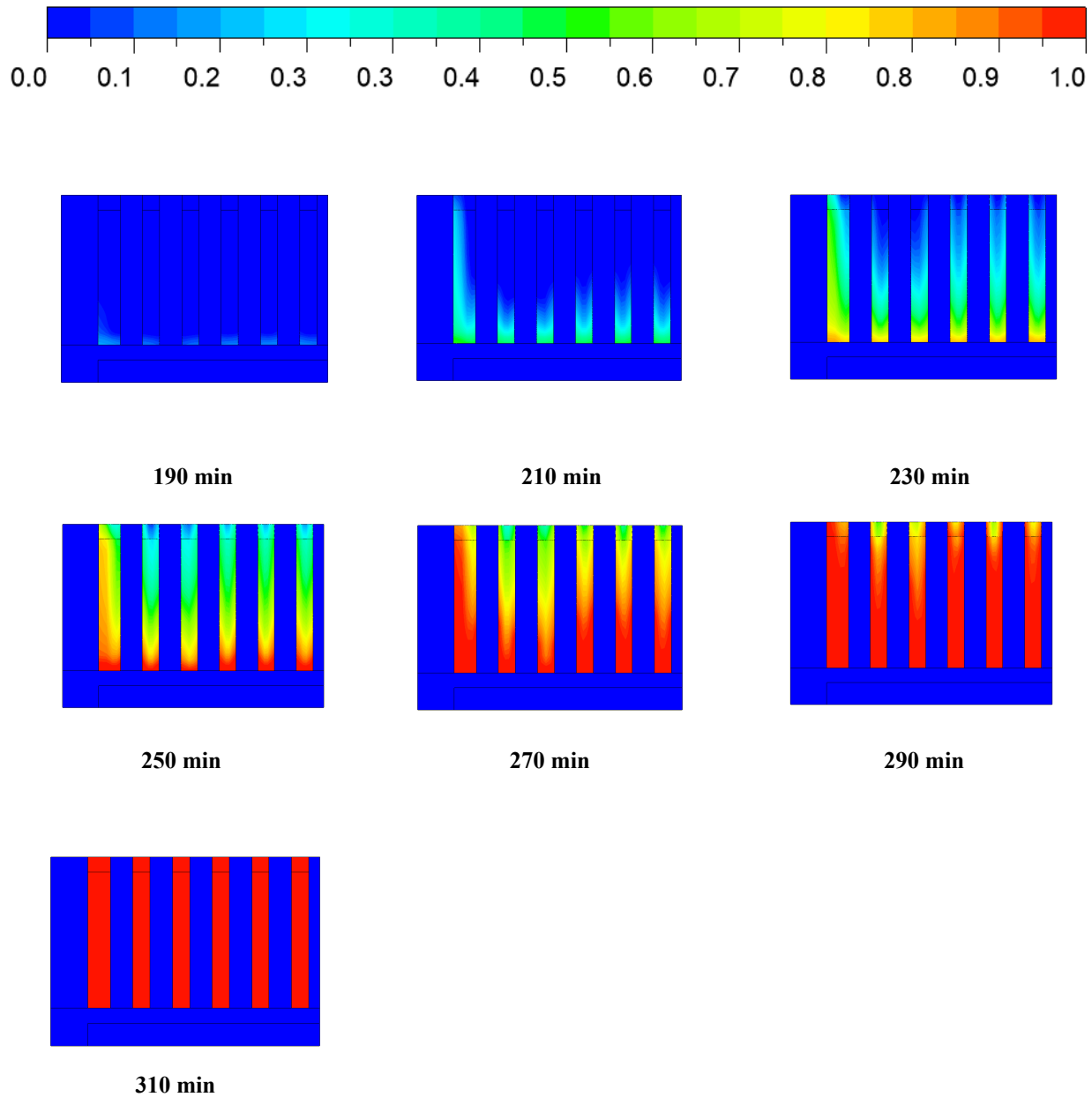
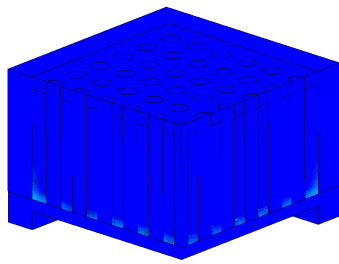
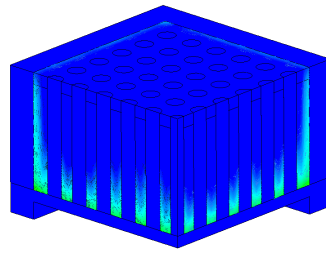


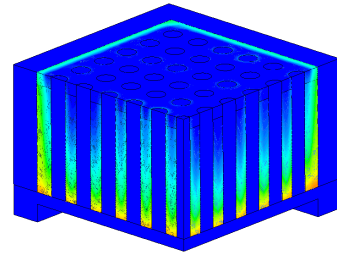
Fig. 2D view of liquid fraction contours at various time periods at power level of 30 W.



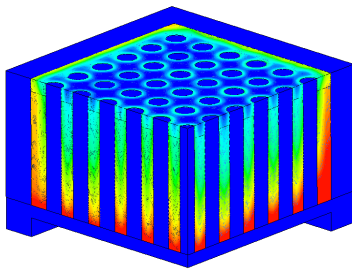
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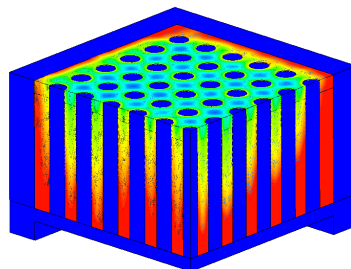
210 min



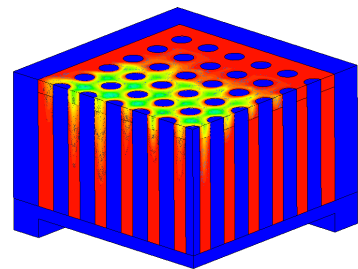
230 min



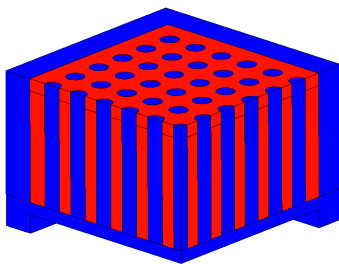
250 min



270 min



290 min



310 min

Fig. 3D view of liquid fraction contours at various time periods at power level of 30 W

4.1.2.2 Temperature Contours at power level of 30 W

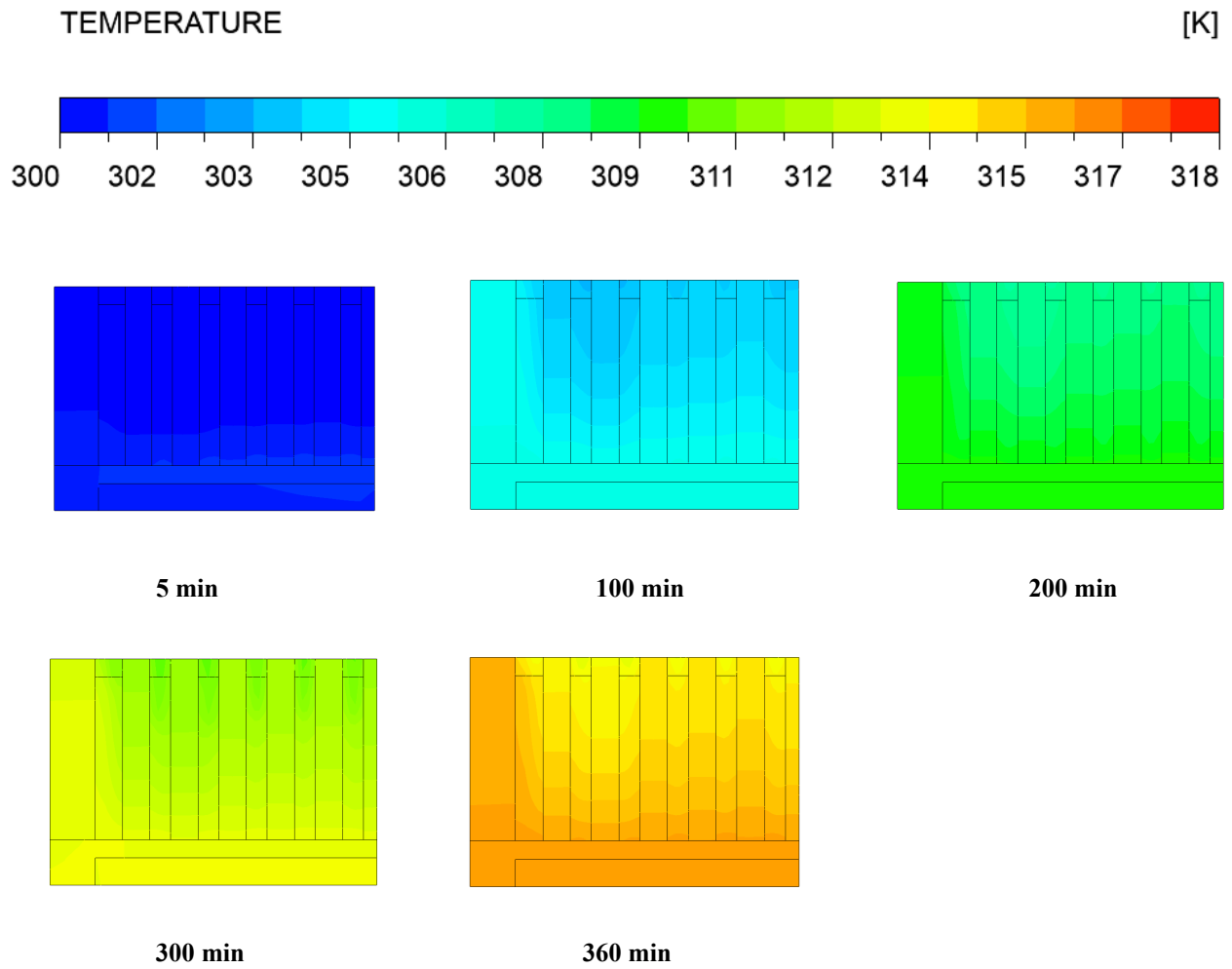


Fig. 2D view of temperature contours at various time periods at power level of 30 W.

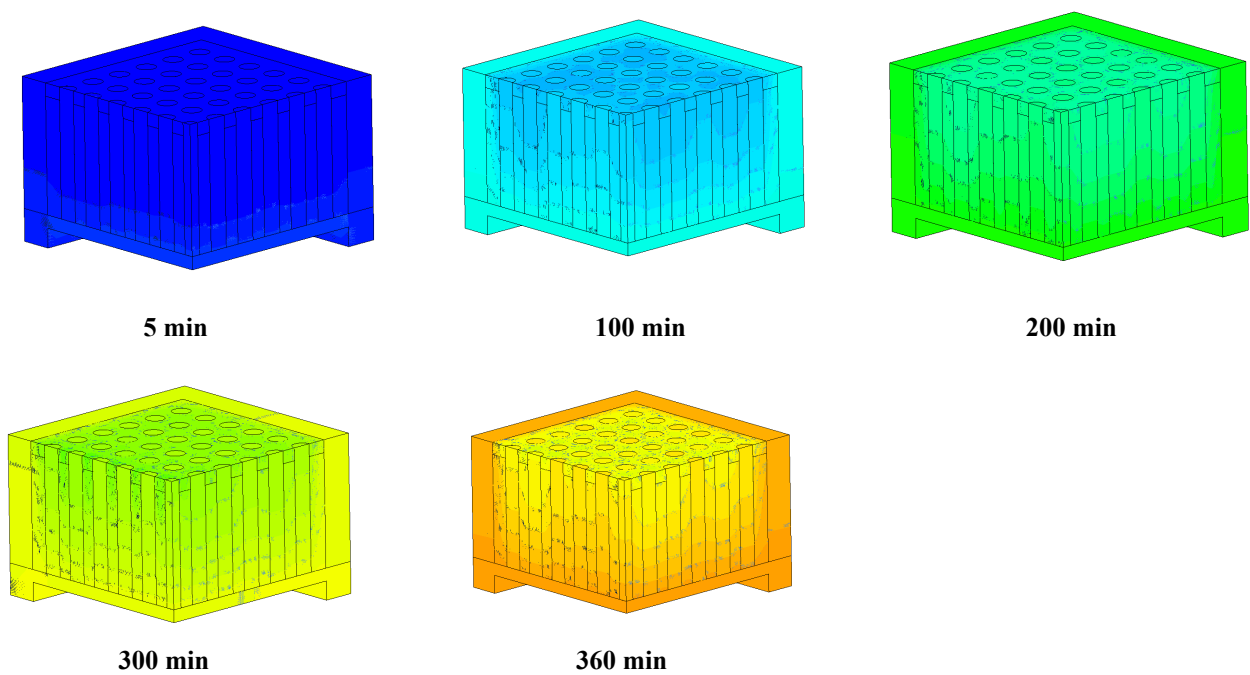


Fig. 3D view of temperature contours at various time periods at power level of 30 W.

4.1.3 THE PIN FIN CONFIGURATION AT 35 W

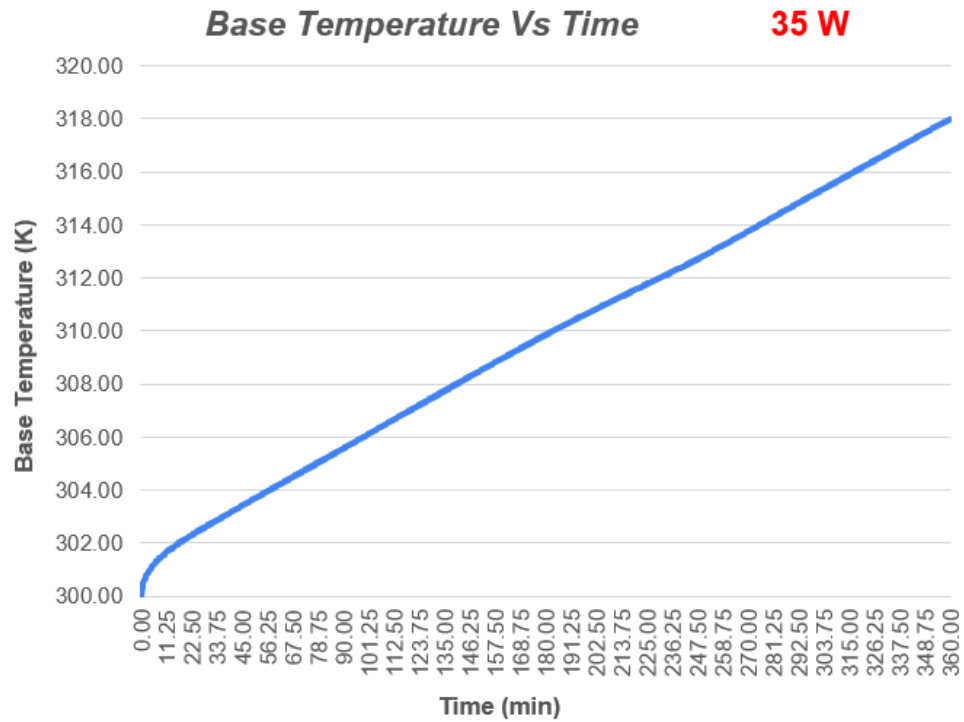


Fig: Base Temperature vs Time graph

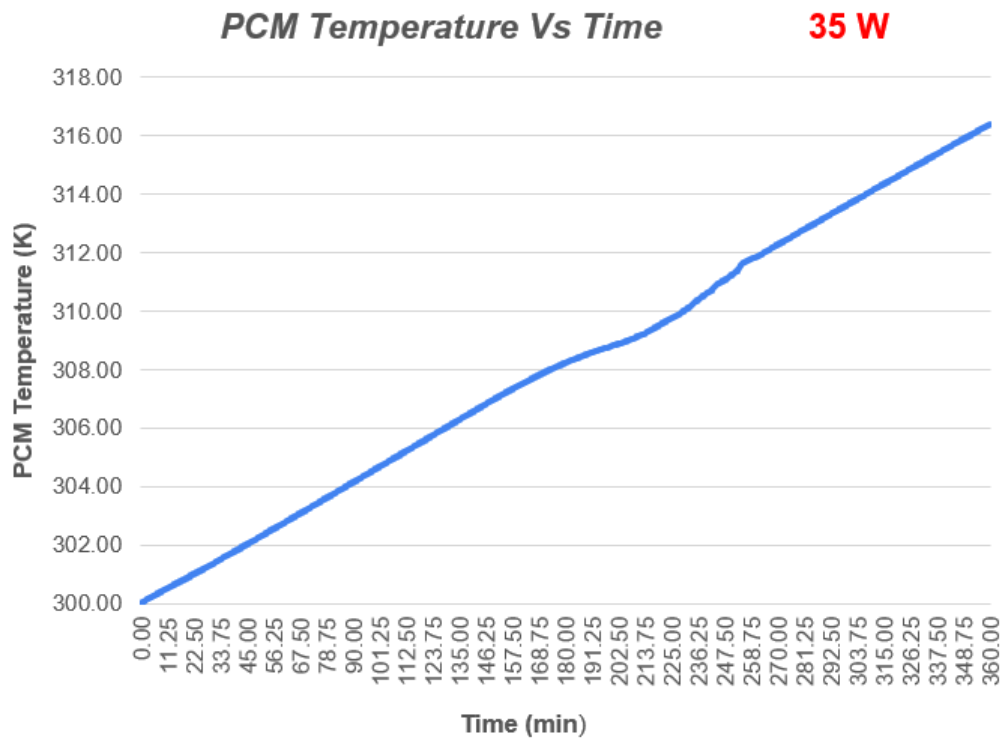


Fig: PCM Temperature vs Time graph

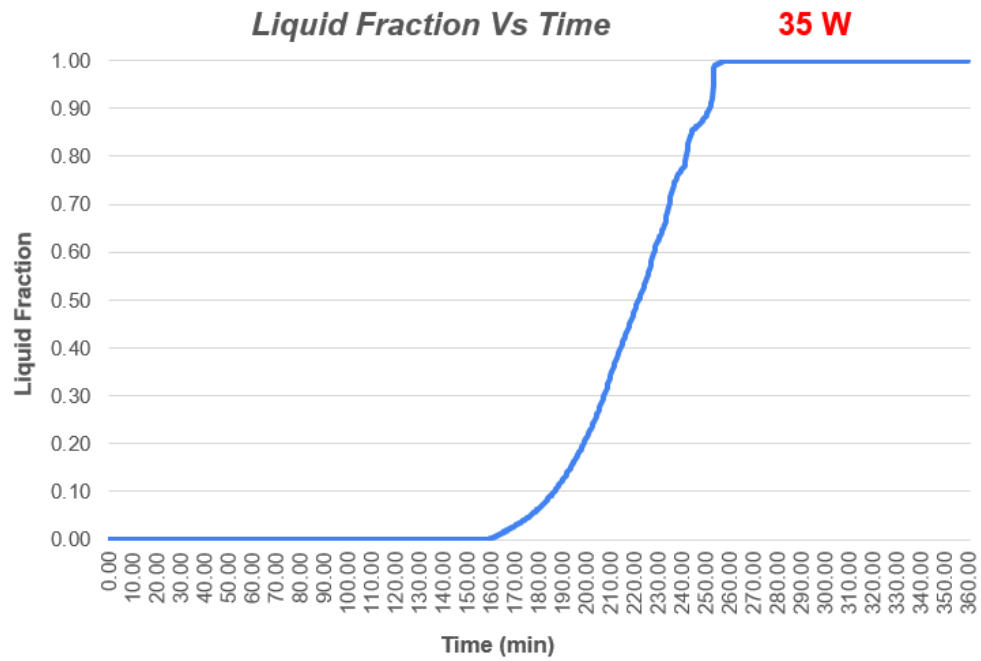
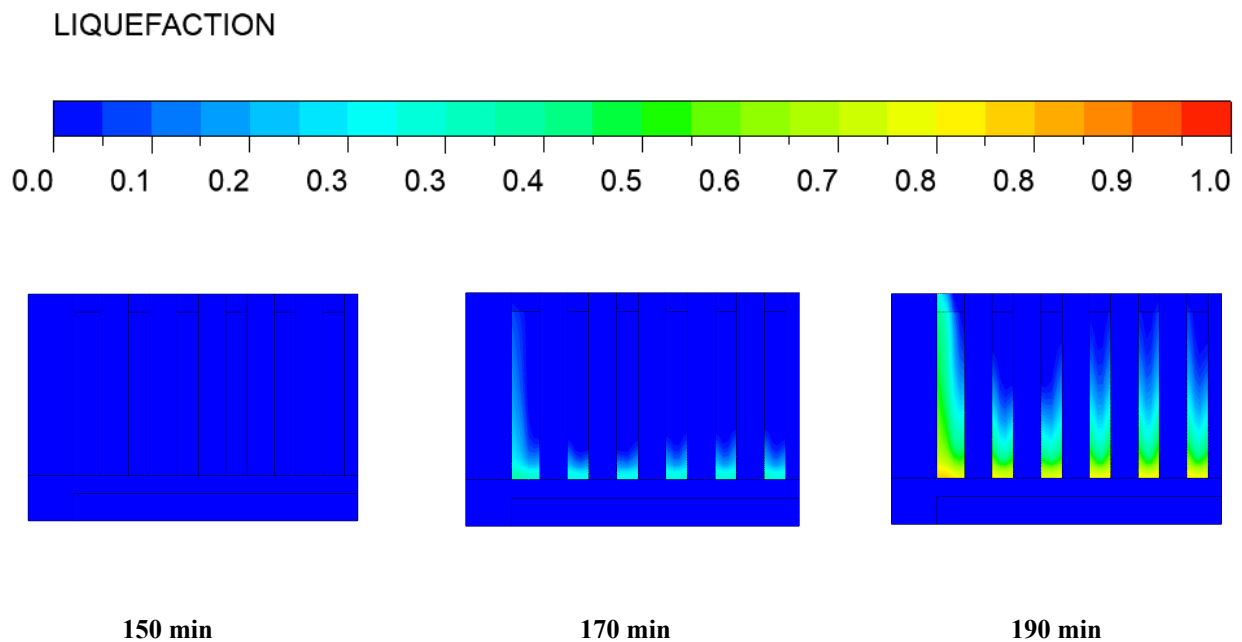
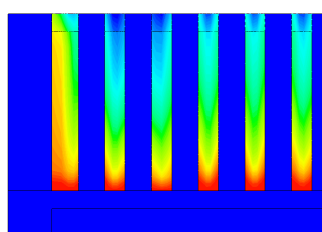


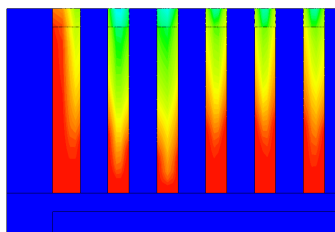
Fig: Liquid fraction vs Flow Time graph

4.1.3.1 Liquid Fraction Contours at power level of 30W

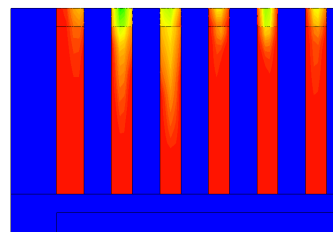




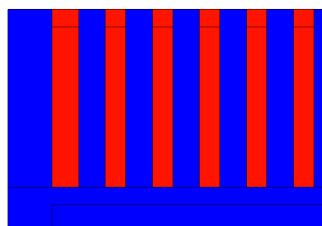
210 min



230 min

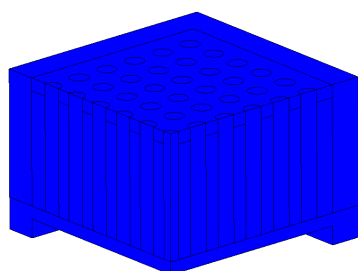


250 min

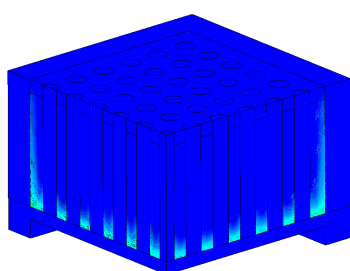


270 min

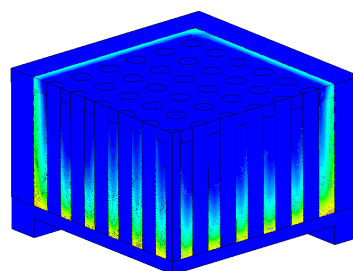
Fig. 2D view of liquid fraction contours at various time periods at power level of 30 W.



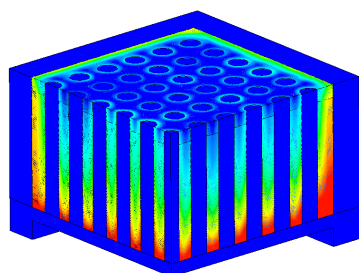
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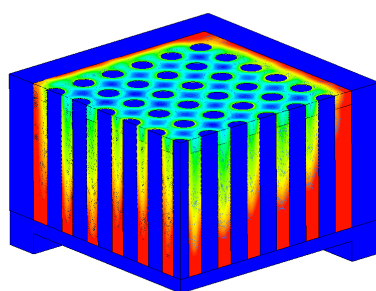
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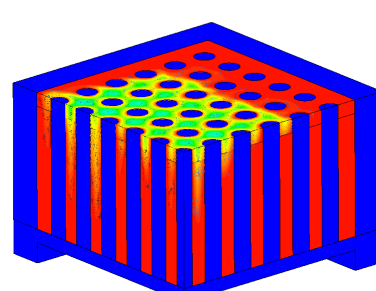
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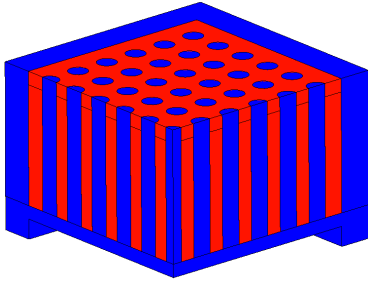
210 min



230 min



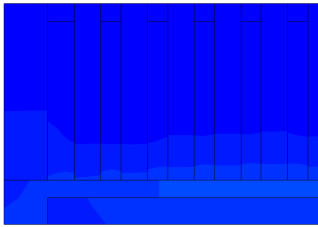
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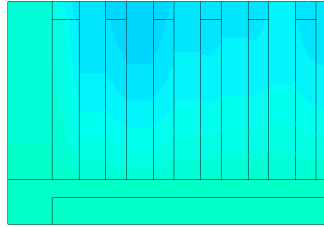
270 min

Fig. 3D view of liquid fraction contours at various time periods at power level of 30 W

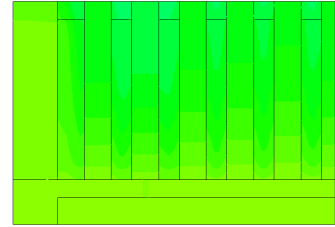
4.1.3.2 Temperature Contours at power level of 30 W



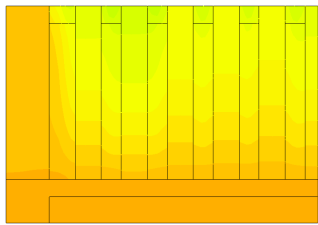
5 min



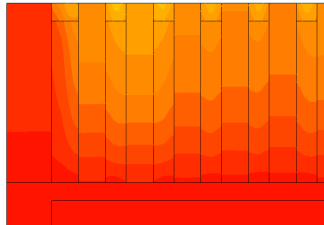
100 min



200 min

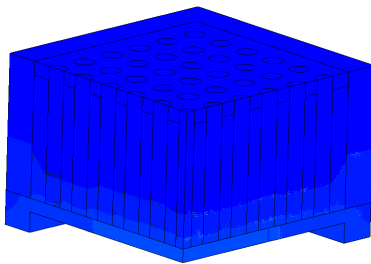


300 min

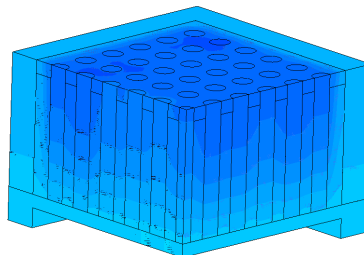


360 min

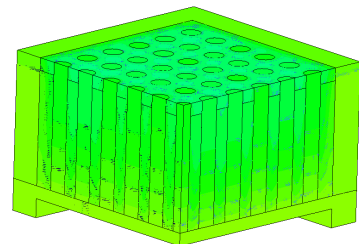
Fig. 2D view of temperature contours at various time periods at power level of 30 W.



5 min



100 min



200 min

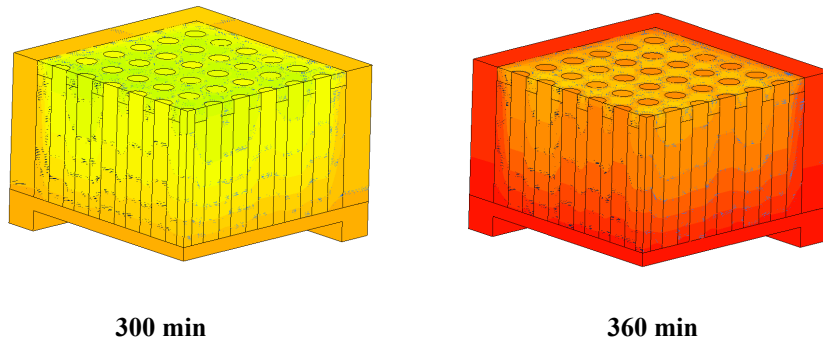


Fig. 3D view of temperature contours at various time periods at power level of 30 W.

4.2 COMPARISON BETWEEN THE PIN FIN CONFIGURATION AT POWER LEVEL OF 25W, 30W AND 35W.

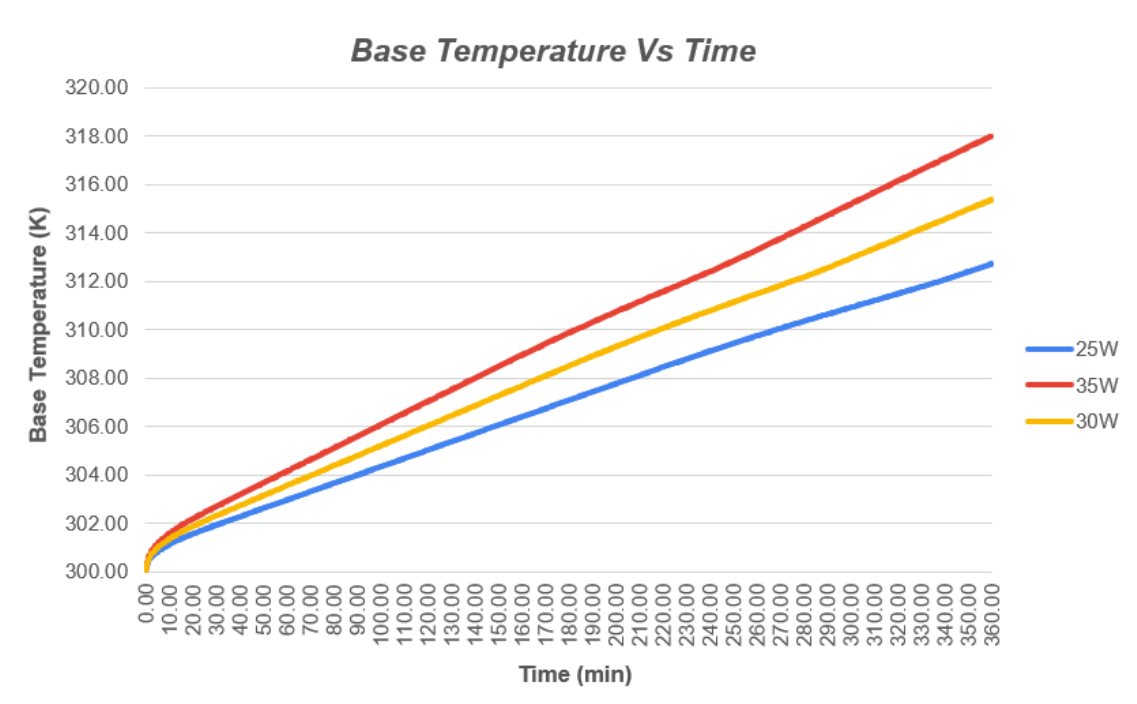


Fig: Base Temperature vs Time graph at different heat inputs

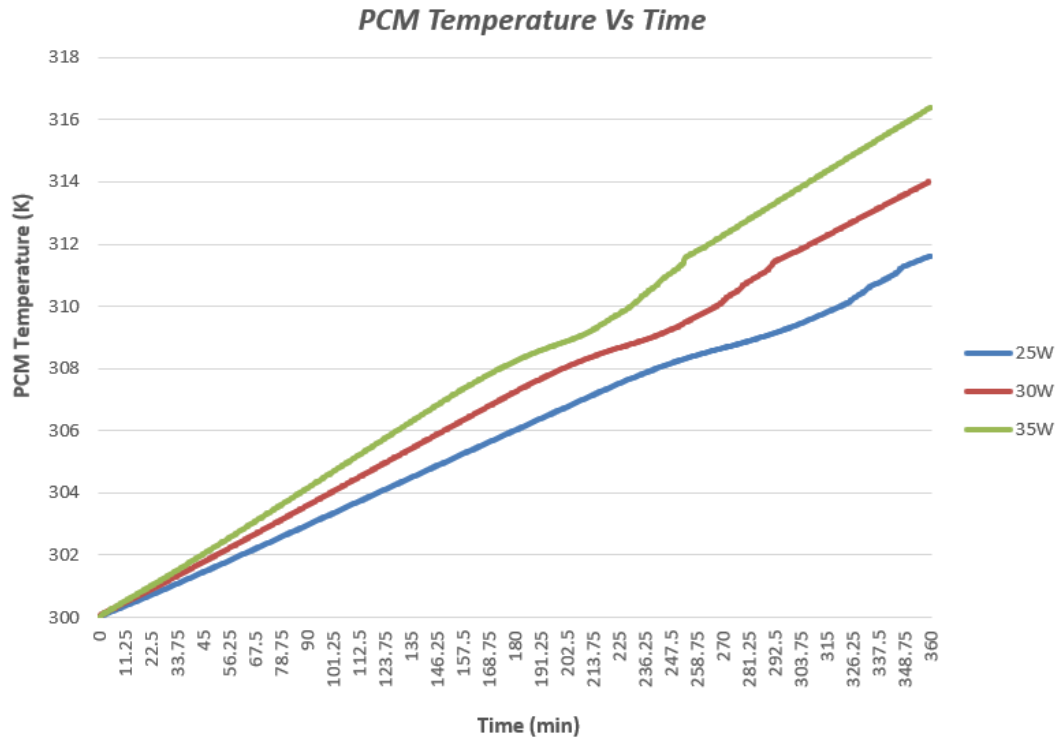


Fig: PCM Temperature vs Time graph at different heat inputs

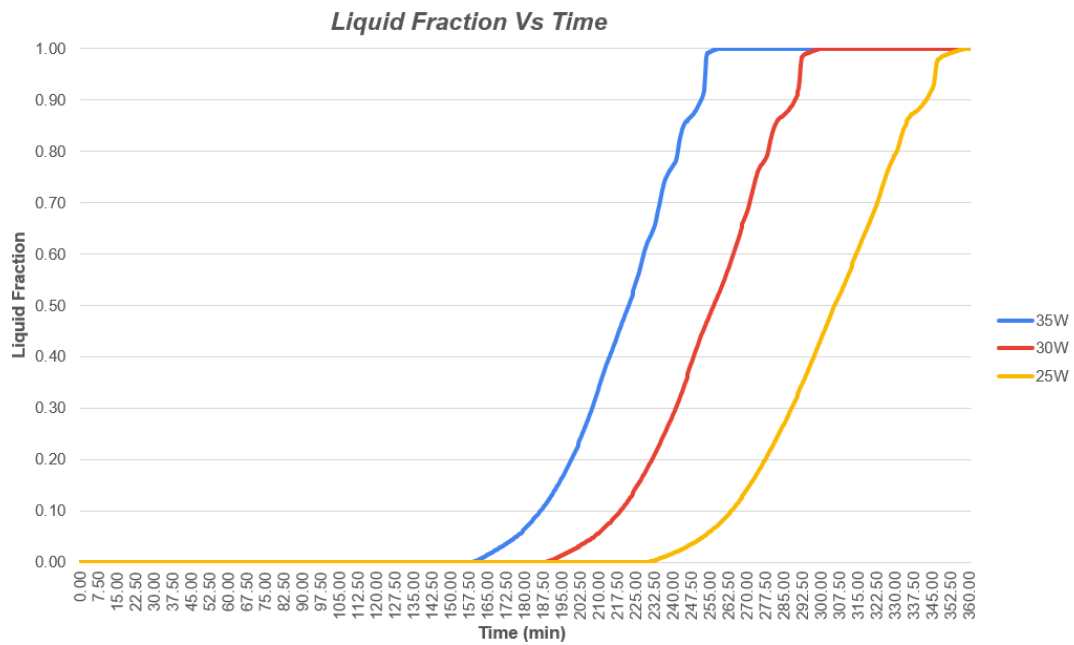
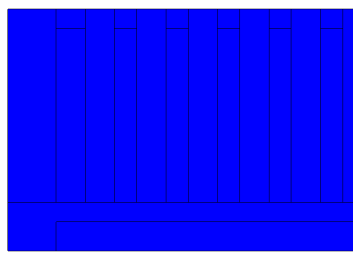
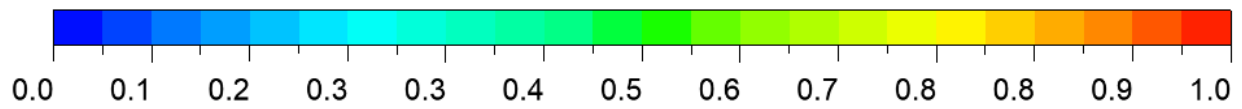


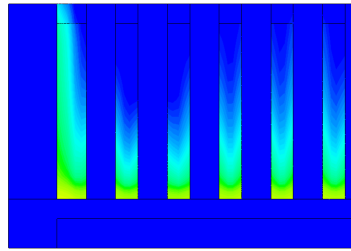
Fig: Liquid fraction vs Time graph at different heat inputs

4.2.1. Comparison of Liquid Fraction Contours at different power level

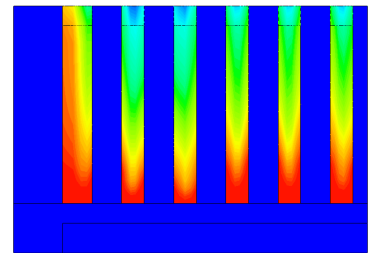
LIQUEFACTION



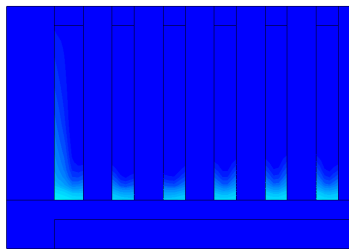
220 min 25W



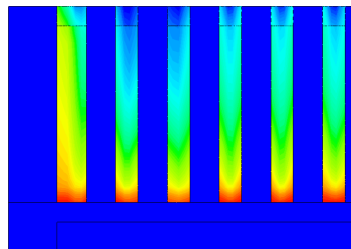
220 min 30W



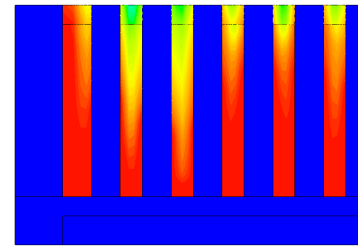
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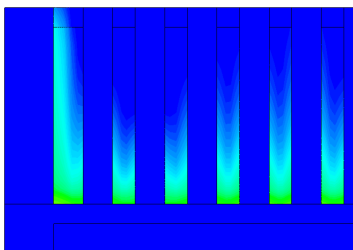
240 min 25W



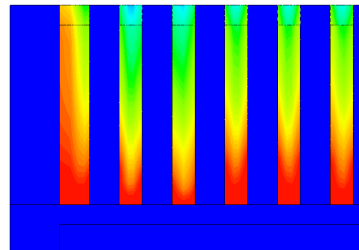
240 min 30W



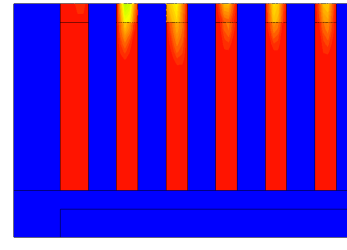
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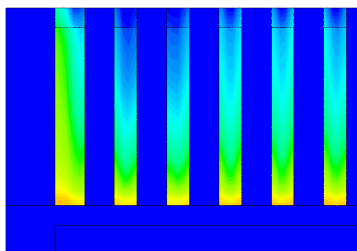
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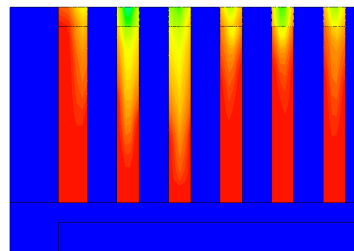
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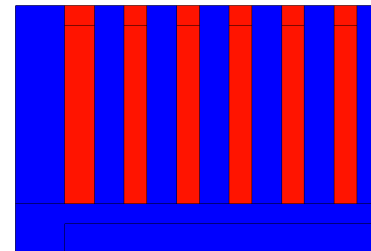
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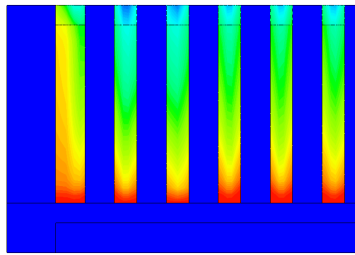
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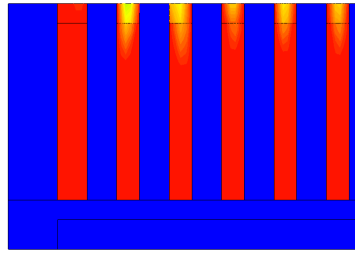
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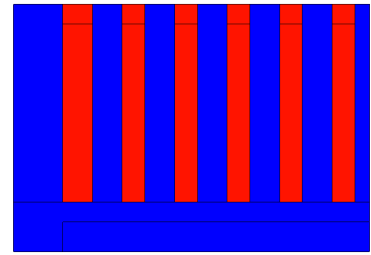
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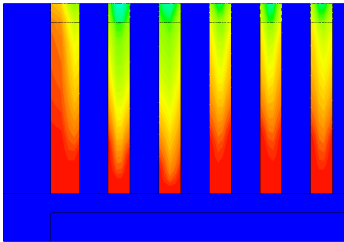
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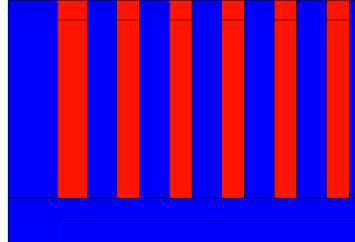
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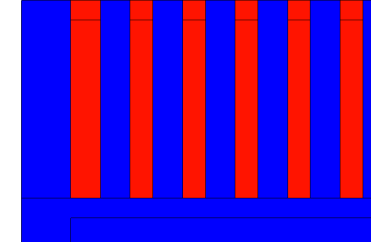
300 min 35W



320 min 25W

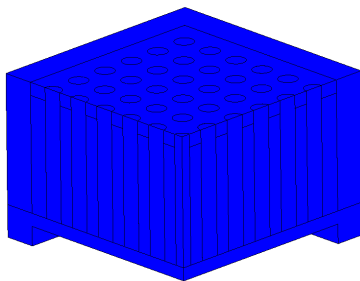


320 min 30W

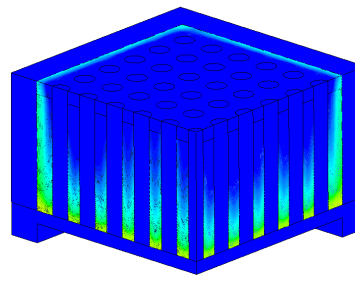


320 min 35W

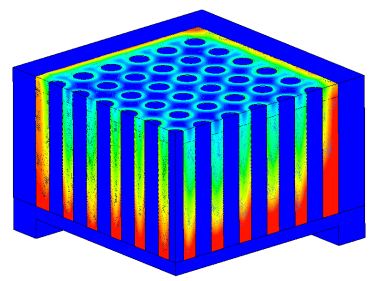
Fig. Comparison of liquid fraction contours at different power levels (2D view)



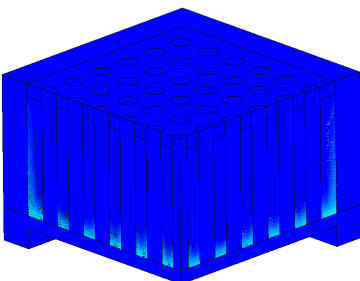
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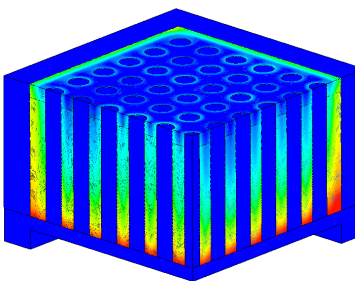
220 min 30W



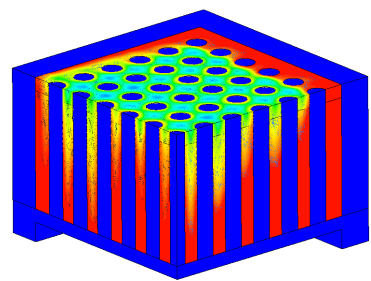
220 min 35W



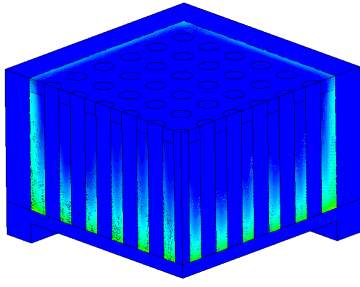
240 min 25W



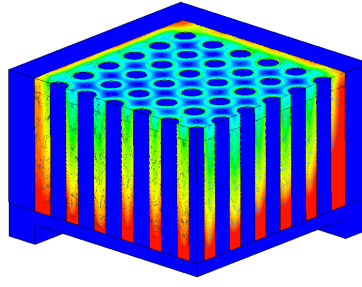
240 min 30W



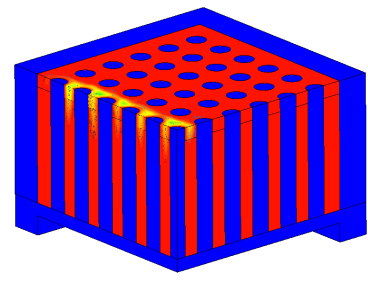
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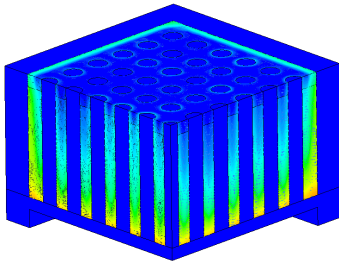
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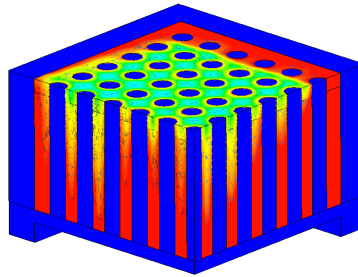
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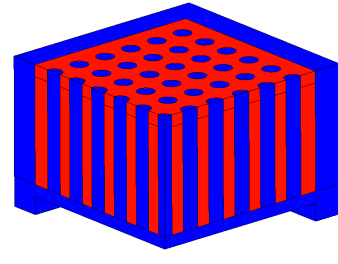
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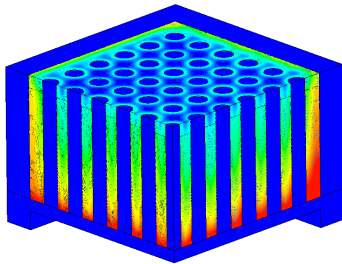
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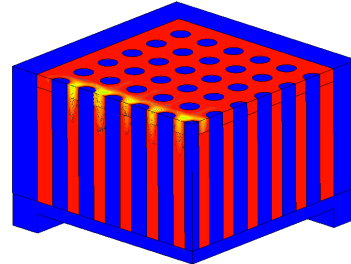
280 min 30W



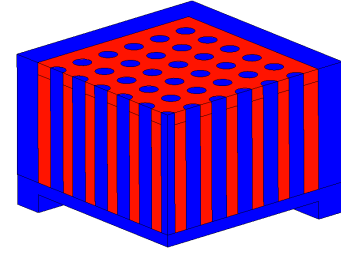
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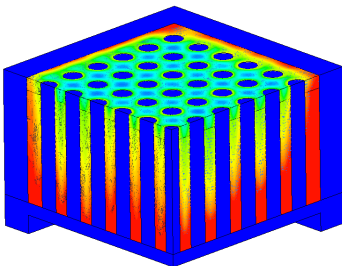
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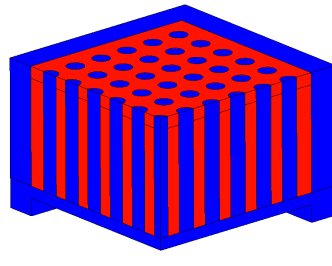
300 min 30W



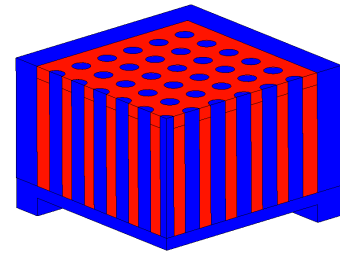
300 min 35W



320 min 25W



320 min 30W



320 min 35W

Fig. Comparison of liquid fraction contours at different power levels (3D view)

4.2.2. Comparison of Temperature Contours at different power level

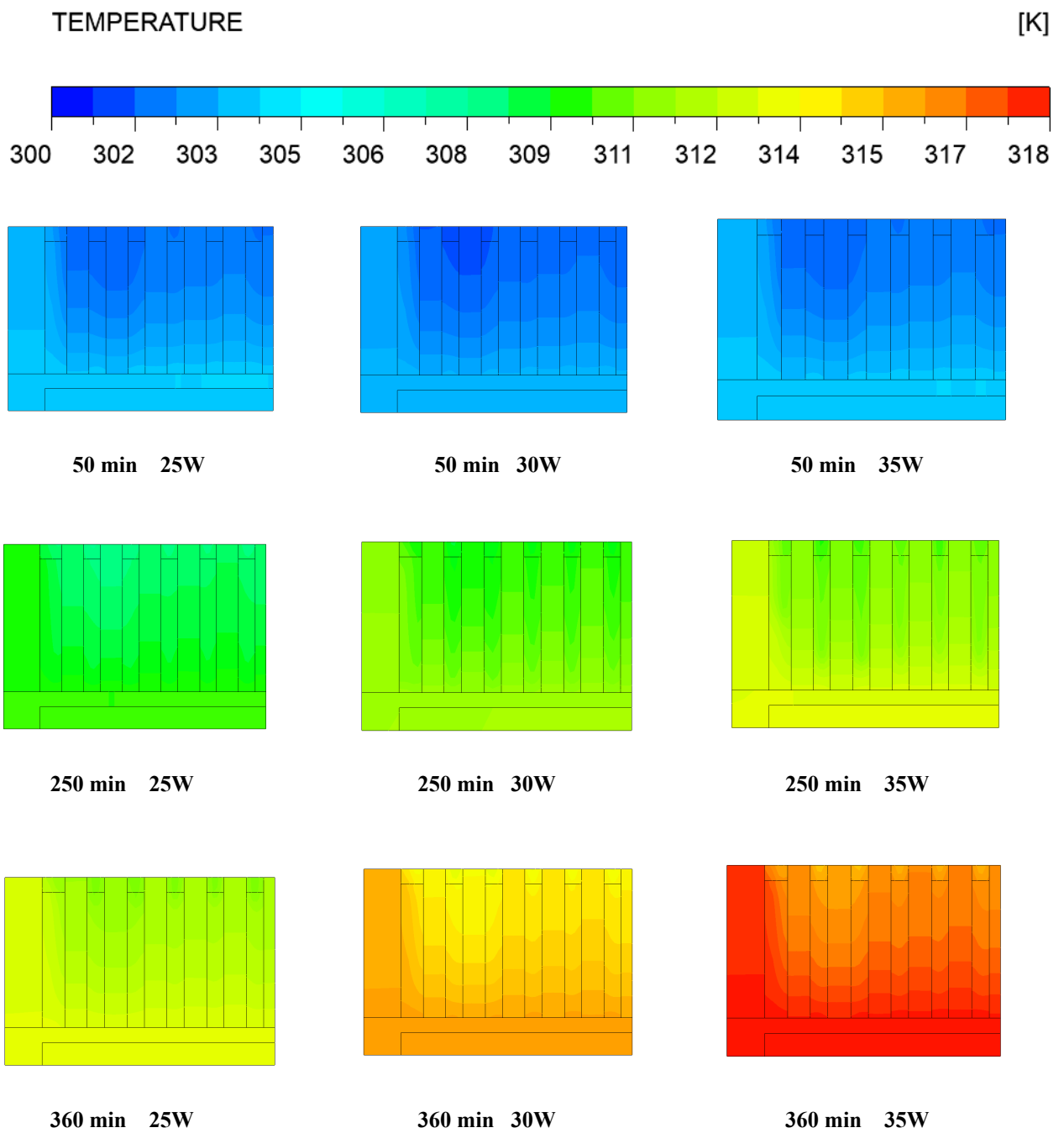
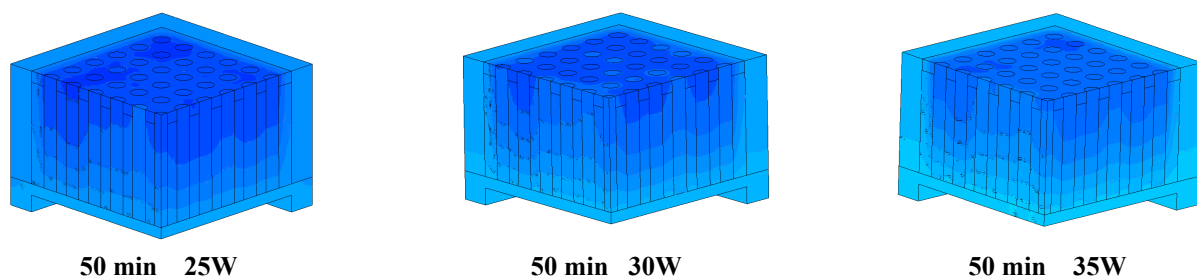
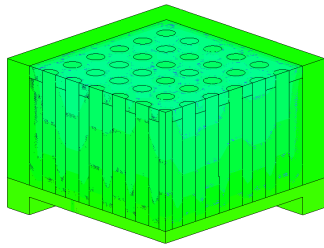
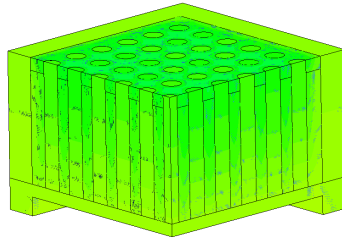


Fig. Comparison of temperature contours at different power levels (2D view)

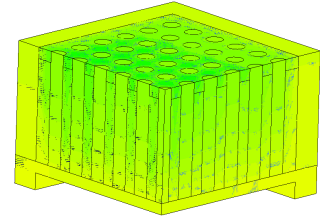




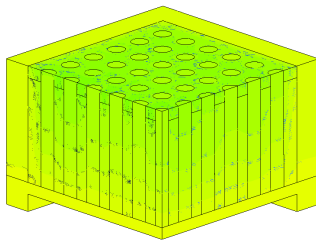
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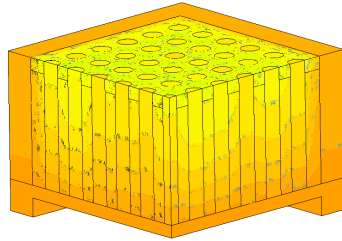
250 min 30W



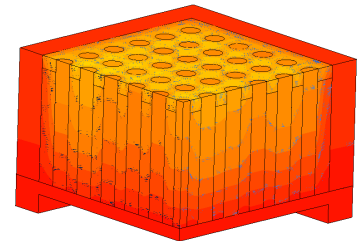
250 min 35W



360 min 25W



360 min 30W



360 min 35W

Chapter 5

5.1. CFD MODEL AND VALIDATION

CFD Model validation is crucial in computational studies for ensuring Accuracy .

Validation ensures that the computational model accurately represents the real-world phenomenon or system being studied. By comparing simulation results with experimental data or established analytical solutions, we can verify the reliability of their models and gain confidence in the accuracy of their predictions.

We have compared the following parameters for validation purpose:

1. Change in temperature: We can observe similar graph curves in the research paper and result generated by us.

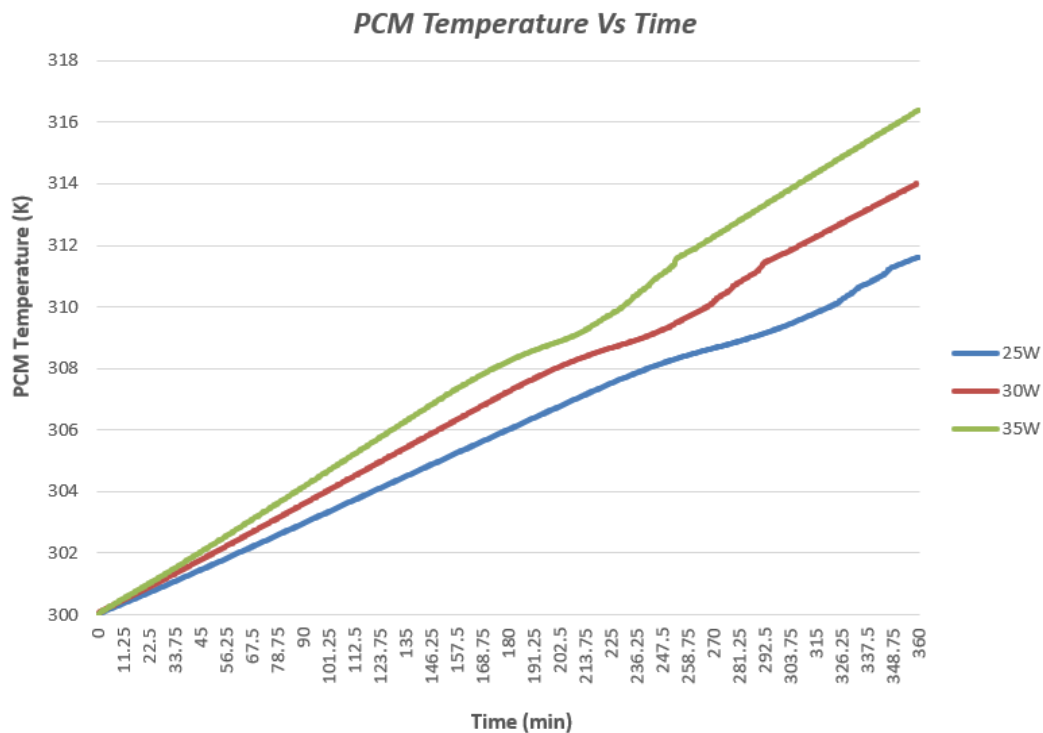


Fig. Combined Graph of Heat Inputs

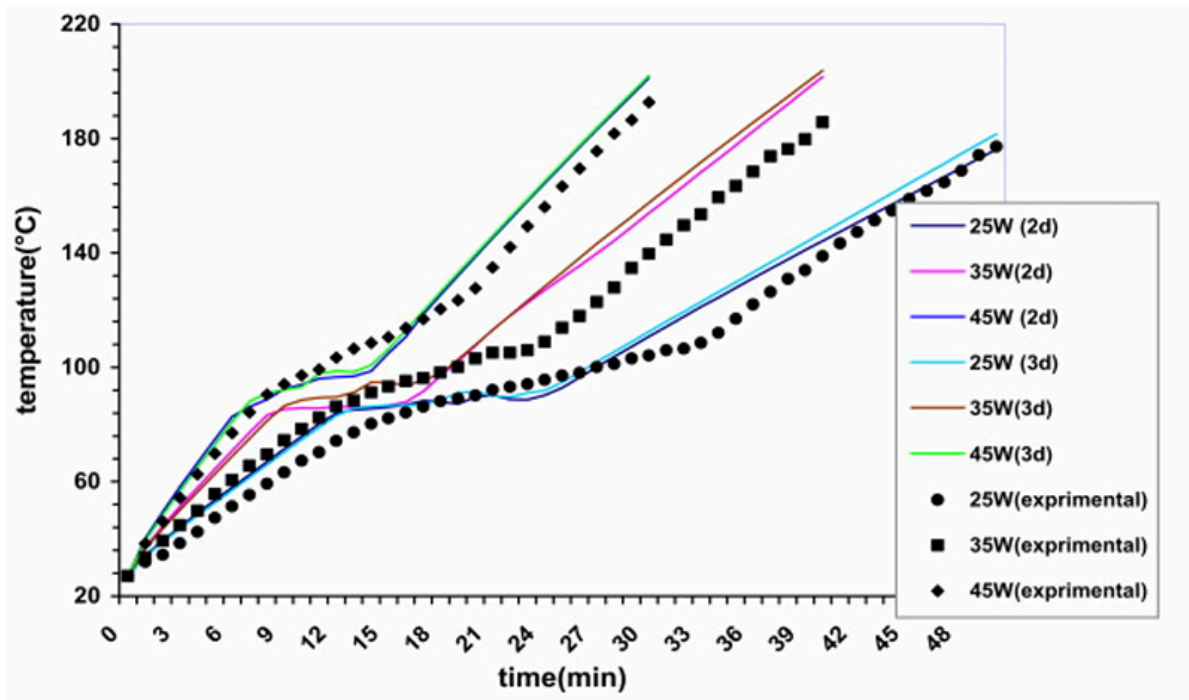


Fig. Combined Graph of Heat Inputs

2. Change in Liquid Fraction: We can observe similar graph curves in the research paper and result generated by us.

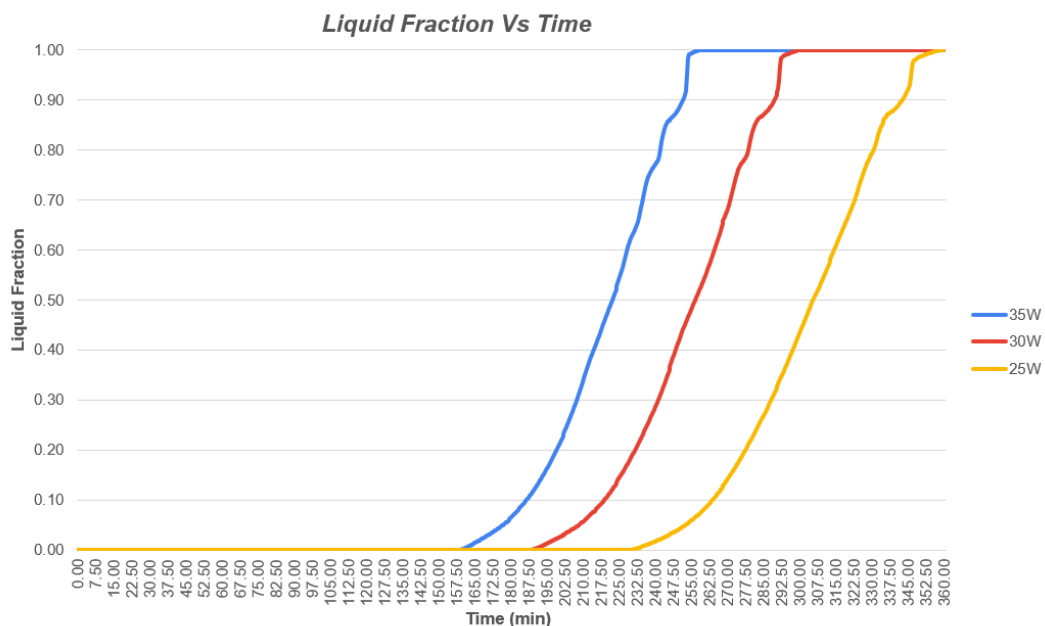


Fig. Combined Graph of Heat Inputs

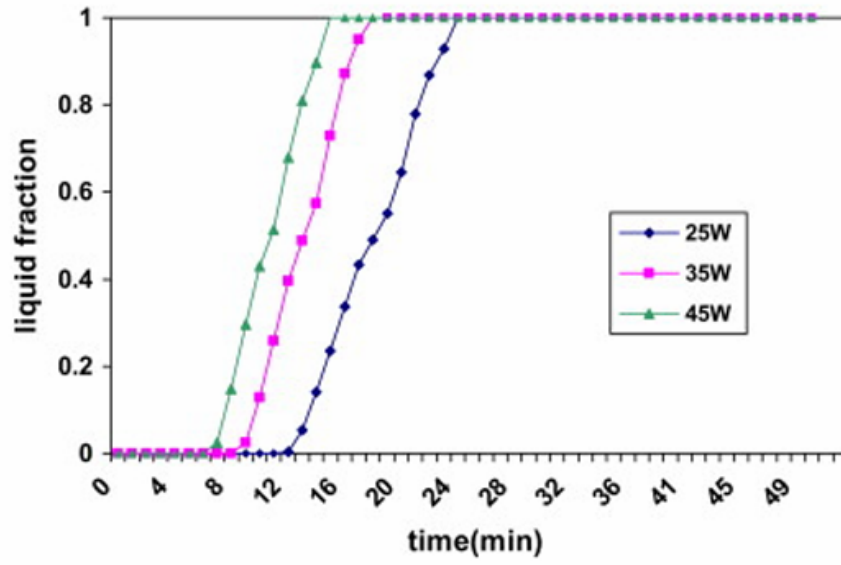


Fig. Combined Graph of Heat Inputs

3. Liquid Fraction Contours: We can observe similar graph curves in the research paper and result generated by us.

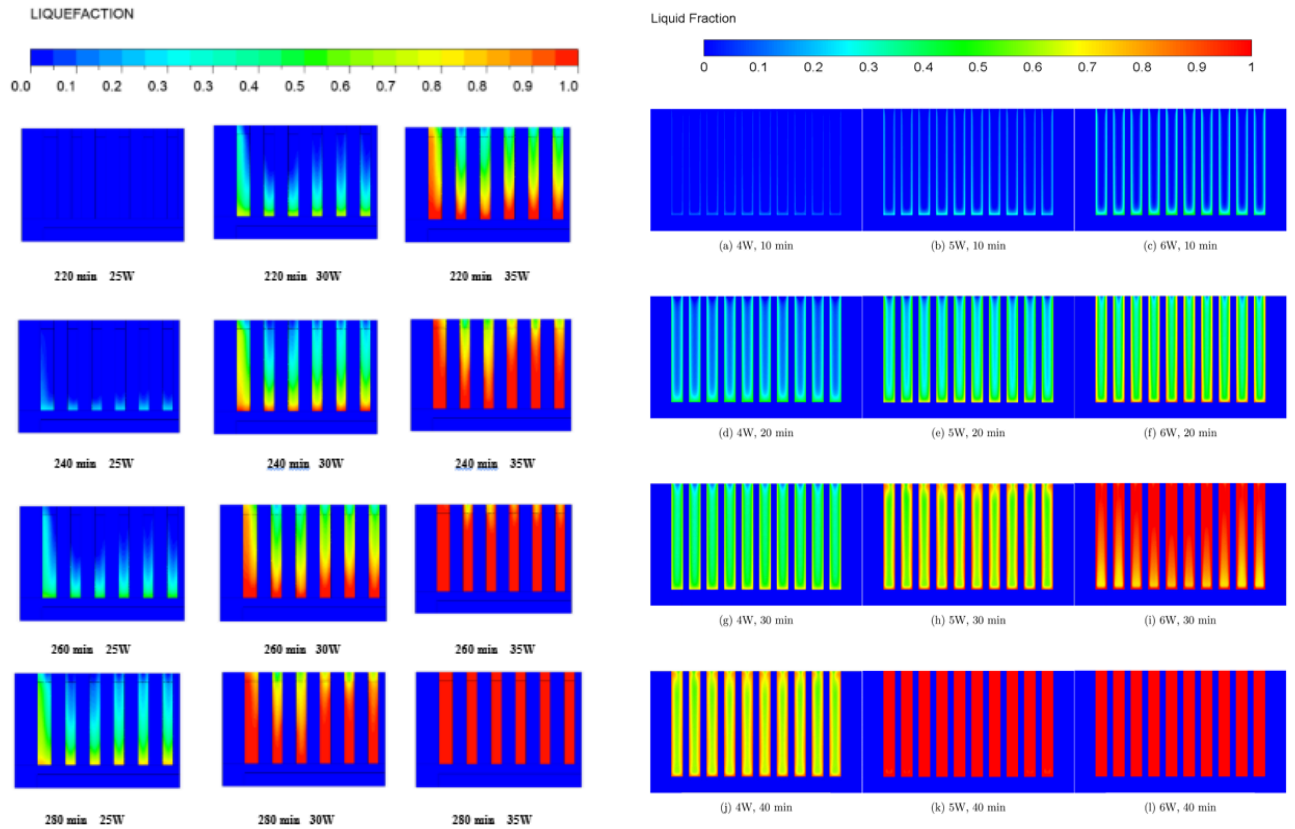


Fig. Combined Graph of liquid fraction

Chapter 6

6.1. CONCLUSIONS

Based on the analysis of CFD-generated plots and renderings across multiple geometries and operating conditions, the following key findings were observed:

1. Phase Change Materials (PCMs) Enhance Thermal Performance

The inclusion of PCMs in the hybrid heat sink improves heat absorption and dissipation through melting/solidification cycles.

This stabilizes temperatures and mitigates overheating risks in electronic components.

2. Heat Input Directly Influences PCM Behavior

Simulations show that higher thermal loads intensify phase change activity, confirming the PCM's ability to handle elevated heat fluxes effectively.

3. PCM Configuration Plays a Critical Role

Optimal PCM placement and volume are essential for maximizing heat dissipation.

Designs promoting uniform temperature distribution and efficient phase transitions delivered superior thermal management.

4. Uniform Temperature Distribution Validated via Contour Plots

The hybrid heat sink demonstrated fewer hot spots and more even temperature gradients than conventional designs, improving component reliability.

5. Robust Performance Across Boundary Conditions

The heat sink maintained consistent performance under diverse simulated real-world conditions, underscoring its practical viability for electronic cooling applications.

REFERENCES

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3. <https://www.sciencedirect.com/science/article/pii/S2214157X21001763>
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5. https://www.academia.edu/25818643/Study_of_Solidification_in_Vertical_Cylindrical_Shells#:~:text=The%20present%20study%20explores%20numerically%20the%20process%20of,%28PCM%29%20in%20cylindrical%20shells%20with%20a%20vertical%20axis.
6. <https://www.sciencedirect.com/science/article/pii/S0017931005002115>
7. <https://www.sciencedirect.com/science/article/pii/S1359431119339833#:~:text=In%20the%20present%20study%2C%20heat%20transfer%20characteristics%20of,was%20utilized%20to%20predict%20transient%20heat%20transfer%20coefficient.>
8. <https://www.sciencedirect.com/science/article/pii/S245190492030041X>

